RAVEN-MCS: Target-Risk-Calibrated Asynchronous Federated Data Completion under Two-Stage Non-Random Selection

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Abstract—**Sparse mobile crowdsensing relies on data completion to reconstruct unobserved spatio-temporal values from a limited number of mobile measurements. Federated and asynchronous learning can keep raw measurements on devices and accommodate heterogeneous computation and communication, but the successfully received updates generally do not represent the desired deployment distribution. A client–unit record must first arise from a nonuniform sensing opportunity and produce a valid observation, after which its local update must complete computation, arrive before the aggregation deadline, and remain sufficiently fresh. Consequently, conventional asynchronous aggregation may optimize the distribution of received updates rather than the prescribed city-level target risk. This paper proposes RAVEN-MCS, a target-risk-calibrated asynchronous federated data-completion framework under opportunity imbalance and two-stage non-random selection. RAVEN-MCS maps non-repeated atomic units into repeatable opportunity strata, combines target-to-opportunity design correction with stabilized Hájek observation weighting, and applies a second inverse-propensity correction to usable update generation. At the end of each asynchronous aggregation window, the server maintains a learning-rate-weighted target-debt process and solves a strongly convex calibration problem that jointly controls instantaneous target-group mismatch, long-term under-representation, corrected client–record composition, update variance, and model staleness. The analysis bounds the practical two-stage reference error, establishes long-term group-mixture alignment under approximate window reachability, and derives a nonconvex target-risk convergence bound determined by the per-window client–record distribution error, local optimization drift, staleness, stochastic variance, and population-calibration residual. An evaluation methodology based on controlled completion benchmarks and trace-consistent mobile sensing data separately measures deployment-target error, raw-arrival-weighted error, distribution alignment, target debt, reachability, robustness, and system overhead.**

Index Terms—**Mobile crowdsensing, sparse data completion, federated learning, asynchronous aggregation, non-random selection, importance weighting, target-risk calibration.**

I. Introduction

Mobile crowdsensing (MCS) exploits the sensing, computation, and communication capabilities of pervasive mobile devices to construct large-scale observations of physical and urban environments [1], [2]. It has enabled applications such as environmental monitoring, traffic-state estimation, public-safety assessment, and urban resource management. However, collecting dense observations over every spatial region and time interval is often prohibitively expensive because of limited sensing budgets, uncertain user mobility, device availability, and communication constraints. Sparse mobile crowdsensing (SMCS) addresses this difficulty by collecting measurements from only part of the spatio-temporal field and inferring the remaining values through data completion [3], [4].

Substantial progress has been made in improving SMCS completion. Matrix-completion methods exploit low-rank structure to recover unobserved values, while graph neural networks, attention mechanisms, and temporal representation models capture more complex spatial and temporal dependencies [3]–[5]. Recent studies have further examined whether the computational cost of completion is justified by its benefit to downstream sensing tasks [6]. Nevertheless, most data-completion methods require the platform to collect or centrally process users' location-dependent sensing records, which may reveal sensitive mobility and behavioral information.

Federated learning (FL) provides a natural means of keeping raw records on mobile devices while learning a shared model through locally computed updates [7]. Asynchronous FL further reduces synchronization delays by allowing clients with heterogeneous computation and communication capabilities to return updates at different times [8]. Following this direction, federated distributed completion has been introduced into SMCS by combining local matrix-factorization models, time-aware completion, and asynchronous parameter aggregation [9]. Such a framework substantially improves the deployability of distributed completion under sparse and irregular local data. However, retaining raw measurements at clients does not by itself provide a cryptographic or differential-privacy guarantee, and it does not resolve the statistical bias caused by the way local records and model updates enter federated training.

The central problem considered in this paper is that the data received by an asynchronous SMCS platform are generally not a random sample from the deployment target. A client–unit record must first appear in the client's sensing opportunity set and then produce a valid local observation. Even after local observations are obtained, the corresponding model update may be absent because local computation fails, transmission is interrupted, the update misses the aggregation deadline, or its downloaded model becomes excessively stale. These mechanisms constitute two coupled non-random

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selection stages: *local observation selection* and *usable-update selection*. Moreover, the underlying sensing opportunities are themselves unequally distributed across clients, regions, and recurring temporal contexts.

This issue is more fundamental than ordinary data heterogeneity. Asynchronous aggregation may converge accurately with respect to the distribution of successfully received updates while still performing poorly under the desired city-wide deployment distribution. Related FL studies have shown that partial, nonuniform, or biased client participation can change convergence behavior, increase aggregation variance, and move the learned solution away from the full-participation objective [10]–[12]. However, these methods mainly study client-level participation or sampling. They do not jointly correct the preceding non-random observation process, the recurring client–spatio-temporal opportunity distribution, and the subsequent computation–delivery–expiry process of asynchronous SMCS.

Importance weighting provides a general statistical principle for aligning training and target distributions under identifiable distribution shift [13]. Directly applying an inverse propensity weight to RAVEN-MCS is nevertheless insufficient. First, absolute spatio-temporal units generally occur only once, so their opportunity probabilities cannot be reliably estimated through per-unit historical counts. Second, multiplying two inverse selection probabilities can create highly concentrated local and server weights. Third, a finite aggregation window may not contain all target groups, making exact target matching infeasible in an individual window. Fourth, controlling only the final long-term group proportion does not ensure that each model update approximates the target-risk gradient. A practical solution must therefore combine identifiable two-stage correction with weight stabilization, instantaneous calibration, long-term coverage control, variance regulation, and stale-update management.

To address these challenges, we propose *RAVEN-MCS*, a target-risk-calibrated asynchronous federated data completion framework under opportunity imbalance and two-stage non-random selection. RAVEN-MCS first maps atomic spatio-temporal units into repeatable opportunity strata, allowing the platform to estimate recurring client–context opportunity distributions without relying on non-repeated absolute time slots. Each client combines the resulting target-to-opportunity design ratio with a lagged observation propensity and constructs a stabilized Hájek-weighted local completion objective. The server registers all local training attempts and estimates the probability that each attempt produces an on-time, sufficiently fresh, usable update. A second self-normalized correction then reconstructs a reference client–record composition from the successfully received updates.

RAVEN-MCS does not directly impose this statistical reference as the aggregation vector. Instead, it maintains a learning-rate-weighted target-debt process that records accumulated group under-representation and solves a strongly convex calibration problem at the end of each asynchronous aggregation window. The calibration jointly controls accumulated target debt, instantaneous group mismatch, deviation from the corrected client–record reference, update variance, and model staleness. This separation allows complementary opportunities from different windows to achieve long-term target coverage while directly limiting the distributional bias of each global update.

The main contributions of this work are summarized as follows.

- We formulate target-risk misalignment in asynchronous federated data completion by distinguishing the prescribed public deployment distribution from the distribution induced by sensing opportunities, valid observations, and usable model updates. The model accommodates recurring opportunity strata, slowly varying opportunity distributions, heterogeneous client measurements, and a measurable population-calibration residual.
- We develop RAVEN-MCS, which integrates lagged target-to-opportunity design correction, stabilized local Hájek learning, usable-update propensity correction, and windowed asynchronous aggregation. Its online calibration simultaneously controls instantaneous target mismatch, long-term group debt, corrected client composition, update variance, and staleness.
- We establish a theoretical framework that separately characterizes statistical correction and dynamic optimization. The analysis bounds practical two-stage reference error, relates target debt to the long-term group mixture, characterizes the residual caused by limited window reachability, and derives a nonconvex target-risk convergence bound governed by the actual per-window client–record discrepancy, local optimization drift, staleness, stochastic variance, and population-calibration error.
- We design an evaluation methodology that separates deployment-target error from raw-arrival-weighted error and measures opportunity-ratio estimation, two-stage correction, client–record alignment, target debt, window reachability, variance, and system overhead under controlled and trace-consistent SMCS settings.

The remainder of this paper is organized as follows. Section III presents the system model and problem formulation. Section IV describes the proposed framework. Section V provides the theoretical analysis. Section VI presents the performance evaluation, and the final section concludes the paper.through query-and-answer mechanisms. However,

II. Related Work

A. Data Completion in Sparse Mobile Crowdsensing

Sparse mobile crowdsensing reduces sensing expenditure by collecting measurements from only part of a spatio-temporal field and inferring the remaining values. Early studies primarily formulated this task as a matrix-completion problem. Xie et al. [3] jointly considered active sensing scheduling and matrix completion, thereby improving recovery quality under a limited sensing budget. Subsequent methods introduced more expressive spatial and temporal models. For example, GAMC combines graph representations and attention mechanisms to address spatial and temporal fractures, in which entire regions or time intervals may be unobserved [4]. Recent work has further moved beyond fixed discrete-time completion. TIME-DMF incorporates irregular physical time intervals into a time-continuous completion model [5], while other studies examine whether the computational cost of data completion is justified by its benefit to downstream sensing tasks [6].

Most of these methods assume that the observed entries used for model training are an adequate representation of the intended sensing field. Their main focus is the completion architecture, the exploitation of spatio-temporal correlation, or the selection of sensing locations. Consequently, they generally do not distinguish the deployment target distribution from the distribution induced by users' mobility, sensing availability, and successful data reporting.

Federated data completion has recently been introduced to avoid centralizing raw spatio-temporal measurements. Wang et al. [9] proposed a federated matrix-factorization framework with time-aware local completion and asynchronous model aggregation. This work provides the most direct foundation for distributed SMCS completion and demonstrates how irregular local observations can be processed without transmitting raw records. Nevertheless, its aggregation mechanism primarily addresses local data sparsity and asynchronous arrival. It does not explicitly correct the unequal probability that a client–unit opportunity becomes a valid observation, nor the subsequent probability that its local update completes, arrives before the aggregation deadline, and remains sufficiently fresh. Therefore, the distribution represented by successfully delivered updates may still differ from the prescribed deployment target.

B. Asynchronous Federated Learning with Nonuniform Participation

Federated averaging trains a shared model by repeatedly aggregating updates computed from decentralized client data [7]. To reduce waiting time caused by heterogeneous devices, asynchronous federated optimization allows clients to communicate at different times and analyzes the effects of delayed updates [8], [14]. These studies establish the feasibility of federated optimization under heterogeneous computation and communication, but typically treat client participation as uniform, sufficiently regular, or exogenously given.

A growing body of research has shown that partial and nonuniform participation can affect both variance and the solution reached by federated learning. Clustered sampling improves client representativeness while reducing the variance of stochastic aggregation weights [10]. FedVARP reduces the optimization variance caused by partial participation by maintaining surrogate updates for absent clients [12]. Cho et al. [11] analyze biased client selection and show that deliberately favoring particular clients may accelerate optimization while changing the limiting solution. Wang and Ji [15] provide a unified analysis covering a broad class of arbitrary participation patterns. More recently, correlated and nonuniform availability has been modeled explicitly, together with mechanisms that debias the resulting federated updates [16].

These methods mainly address participation at the client level. They assume that a participating client's local dataset or gradient is already available and then study which clients communicate or how their updates should be aggregated. In SMCS, however, selection begins before local training: mobility and sensing conditions determine the client–spatio-temporal opportunity stream, and only some opportunities yield valid measurements. Moreover, computation failure, communication delay, deadline expiry, and model staleness jointly determine whether a locally generated update is usable. Correcting only the final client-participation process therefore cannot generally reconstruct the desired client–record distribution.

Target-oriented federated learning is also related to our objective. On-Demand FL selects clients whose aggregate class distribution approximates a requested target distribution [17]. Its target-matching perspective is valuable, but it is designed for client selection using inferred class proportions. RAVEN-MCS instead considers a continuously generated spatio-temporal target, corrects two sequential non-random selection stages, and calibrates the updates that actually become available. It further distinguishes instantaneous target-gradient alignment from long-term target-group representation.

C. Importance Weighting and Target-Distribution Calibration

Inverse-probability weighting provides a classical means of recovering population quantities from samples drawn with unequal inclusion probabilities. The Horvitz–Thompson estimator establishes unbiased estimation when the inclusion probabilities are known [18]. Self-normalized, or Hájek-type, estimators replace population totals with ratio-normalized weighted sums and are widely used to improve finite-sample stability [19].

In statistical learning, importance weighting has been extensively studied for covariate shift, where the training and target covariate distributions differ but the conditional outcome mechanism remains stable. Importance-weighted validation and risk estimation can align model selection with the target distribution [13]. Density-ratio estimation methods, such as least-squares importance fitting, estimate the target-to-source ratio directly without separately estimating the two densities [20]. These methods provide the statistical basis for correcting an identifiable distribution shift, but large inverse weights may substantially increase estimator variance, particularly when target support is weakly represented in the source sample.

RAVEN-MCS builds on these principles but addresses a different sequential and federated setting. First, absolute spatio-temporal units usually do not recur, so RAVEN-MCS estimates opportunity ratios over repeatable client–context strata rather than through per-unit historical frequencies. Second, it combines target-to-opportunity weighting with a local observation propensity and a separate usable-update propensity, thereby correcting two distinct selection stages. Third, it uses flooring, clipping, Hájek normalization, effective sample size, and a lagged variance proxy to control the instability of the resulting weights. Finally, because the target mixture may be infeasible in a single asynchronous aggregation window, RAVEN-MCS combines instantaneous group calibration with learning-rate-weighted target debt. Hence, unlike existing completion, participation, or importance- weighting methods considered separately, RAVEN-MCS jointly addresses record-level selection, update-level selection, per-window target-risk alignment, and long-term target coverage in asynchronous federated SMCS.

III. System Model and Problem Formulation

A. Public Spatio-Temporal Completion Target

We consider a mobile crowdsensing system consisting of a platform server and a set of mobile clients

$$\mathcal{K} = \{1, \dots, K\}. \tag{1}$$

The sensing region is partitioned into N spatial cells, and the sensing horizon is divided into T time slots. An atomic spatio-temporal completion unit is

$$i = (n, t) \in \mathcal{I}, \qquad \mathcal{I} = \{1, \dots, N\} \times \{1, \dots, T\}. \tag{2}$$

Let $Y_i \in \mathbb{R}$ denote the latent public sensing value associated with unit i.

The platform maintains a differentiable completion model $f_{\boldsymbol{\theta}}$ with parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$. Given a deployment context $\mathcal{C}_i^{\mathrm{dep}}$, the public city-level prediction is

$$\widehat{Y}_i = f_{\boldsymbol{\theta}}\left(i; \mathcal{C}_i^{\mathrm{dep}}\right). \tag{3}$$

The context may include a sparse observation matrix, its observation mask, timestamps, and admissible spatial and temporal features. Its test-time distribution is fixed independently of the learning method. Therefore, all compared methods are evaluated under the same public completion task and deployment context.

Atomic units are mapped into target calibration groups through

$$h : \mathcal{I} \to \mathcal{G}, \qquad \mathcal{G} = \{1, \dots, G\}, \tag{4}$$

where

$$\mathcal{I}_g = \{i \in \mathcal{I} : h(i) = g\}. \tag{5}$$

The application specifies a target group distribution

$$\boldsymbol{\mu} = [\mu_1, \dots, \mu_G]^{\mathsf{T}}, \qquad \mu_g \geq 0, \qquad \sum_{g=1}^{G} \mu_g = 1, \tag{6}$$

and fixed within-group unit weights

$$\nu_{i|g}^{\mathrm{tar}} \geq 0, \qquad \sum_{i \in \mathcal{I}_g} \nu_{i|g}^{\mathrm{tar}} = 1. \tag{7}$$

The target mass assigned to atomic unit i is

$$\varpi_i^{\mathrm{tar}} = \mu_{h(i)} \nu_{i|h(i)}^{\mathrm{tar}}, \qquad \sum_{i \in \mathcal{I}} \varpi_i^{\mathrm{tar}} = 1. \tag{8}$$

For uniform city-wide completion, $\mu_g = 1/G$ and $\nu_{i|g}^{\mathrm{tar}} = 1/|\mathcal{I}_g|$. For a service-weighted deployment, these quantities may instead reflect externally specified monitoring priorities.

Using the squared loss, the target group risk is

$$F_g(\boldsymbol{\theta}) = \sum_{i \in \mathcal{I}_g} \nu_{i|g}^{\mathrm{tar}} \mathbb{E}_{\mathcal{C}_i^{\mathrm{dep}}} \left[\frac{1}{2}\left(f_{\boldsymbol{\theta}}(i; \mathcal{C}_i^{\mathrm{dep}}) - Y_i\right)^2\right]. \tag{9}$$

The desired deployment risk is

$$F_{\boldsymbol{\mu}}(\boldsymbol{\theta}) = \sum_{g=1}^{G} \mu_g F_g(\boldsymbol{\theta}) = \sum_{i \in \mathcal{I}} \varpi_i^{\mathrm{tar}} F_i(\boldsymbol{\theta}), \tag{10}$$

where

$$F_i(\boldsymbol{\theta}) = \mathbb{E}_{\mathcal{C}_i^{\mathrm{dep}}} \left[\frac{1}{2}\left(f_{\boldsymbol{\theta}}(i; \mathcal{C}_i^{\mathrm{dep}}) - Y_i\right)^2\right]. \tag{11}$$

B. Repeatable Opportunity Strata

A specific atomic unit $i = (n, t)$ generally appears only once and therefore cannot support reliable estimation of a client–unit opportunity frequency. We introduce a repeatable opportunity-stratum mapping

$$s : \mathcal{I} \to \mathcal{S}, \qquad \mathcal{S} = \{1, \dots, S\}. \tag{12}$$

A stratum may combine spatial region, time-of-day block, day type, task category, and other pre-outcome attributes that recur across multiple sensing windows.

Let

$$\mathcal{I}_s = \{i \in \mathcal{I} : s(i) = s\}. \tag{13}$$

The target mass of stratum s is

$$\Lambda_s^{\mathrm{tar}} = \sum_{i \in \mathcal{I}_s} \varpi_i^{\mathrm{tar}}. \tag{14}$$

For $\Lambda_s^{\mathrm{tar}} > 0$, define the conditional target-unit distribution

$$\nu_{i|s}^{\mathrm{tar}} = \frac{\varpi_i^{\mathrm{tar}}}{\Lambda_s^{\mathrm{tar}}}, \qquad i \in \mathcal{I}_s. \tag{15}$$

Let $\lambda_{k|s}^{\mathrm{tar}}$ be the target client share within stratum s:

$$\lambda_{k|s}^{\mathrm{tar}} \geq 0, \qquad \sum_{k=1}^{K} \lambda_{k|s}^{\mathrm{tar}} = 1. \tag{16}$$

It specifies the client population whose measurements are intended to represent the public field within that stratum.

The target client–stratum mass is

$$\pi_{k,s}^{\mathrm{tar}} = \Lambda_s^{\mathrm{tar}} \lambda_{k|s}^{\mathrm{tar}}, \tag{17}$$

and the corresponding target client–unit mass is

$$\pi_{k,i}^{\mathrm{tar}} = \pi_{k,s(i)}^{\mathrm{tar}} \nu_{i|s(i)}^{\mathrm{tar}}. \tag{18}$$

These masses satisfy

$$\sum_{k=1}^{K} \sum_{s=1}^{S} \pi_{k,s}^{\mathrm{tar}} = \sum_{k=1}^{K} \sum_{i \in \mathcal{I}} \pi_{k,i}^{\mathrm{tar}} = 1. \tag{19}$$

At aggregation window r, let $\pi_{k,s,r}^{\mathrm{opp}}$ denote the pre-observation probability that a randomly selected opportunity belongs to client k and stratum s. Within that stratum, let

$$\nu_{k,r}^{\mathrm{opp}}(i \mid s) \tag{20}$$

denote the conditional opportunity probability of atomic unit i.

The ideal opportunity-design ratio is

$$\boxed{\zeta_{k,r,i}^{\circ} = \frac{\pi_{k,s(i)}^{\mathrm{tar}}}{\pi_{k,s(i),r}^{\mathrm{opp}}} \frac{\nu_{i|s(i)}^{\mathrm{tar}}}{\nu_{k,r}^{\mathrm{opp}}(i \mid s(i))}.} \tag{21}$$

The first factor corrects client–stratum opportunity imbalance. The second factor corrects any remaining unit imbalance within the same stratum.

When the opportunity strata are chosen so that atomic units are conditionally exchangeable,

$$\nu_{k,r}^{\mathrm{opp}}(i \mid s) = \nu_{i|s}^{\mathrm{tar}}, \tag{22}$$

and (21) reduces to

$$\zeta^{\circ}_{k,r,s} = \frac{\pi^{\mathrm{tar}}_{k,s}}{\pi^{\mathrm{opp}}_{k,s,r}}. \tag{23}$$

The opportunity process may be stationary or slowly varying. The practical framework uses only opportunity records completed before window r to estimate the current ratio. The theoretical analysis allows

$$\left|\widehat{\zeta}_{k,r,i} - \zeta^{\circ}_{k,r,i}\right| \le \epsilon^{\zeta}_{r}, \tag{24}$$

which includes finite estimation error and temporal opportunity drift.

C. Client Measurement Heterogeneity

When client k has a sensing opportunity for unit i, its potential measurement is

$$Z^{\star}_{k,i} = Y_i + b_{k,i} + \varepsilon_{k,i}, \tag{25}$$

where $b_{k,i}$ is a client-dependent systematic deviation and $\varepsilon_{k,i}$ is random sensing noise satisfying

$$\mathbb{E}\left[\varepsilon_{k,i} \mid k, i, \mathcal{C}^{\mathrm{dep}}_{i}\right] = 0. \tag{26}$$

The target-population mean deviation for atomic unit i is

$$\bar{b}^{\mathrm{tar}}_{i} = \sum_{k=1}^{K} \lambda^{\mathrm{tar}}_{k|s(i)} b_{k,i}. \tag{27}$$

Exact population calibration corresponds to

$$\bar{b}^{\mathrm{tar}}_{i} = 0. \tag{28}$$

To accommodate finite reference populations and real mobility traces, we allow a nonzero calibration residual and define

$$\Delta_{\mathrm{cal}} = \sum_{i\in\mathcal{I}} \varpi^{\mathrm{tar}}_{i} \left|\bar{b}^{\mathrm{tar}}_{i}\right|. \tag{29}$$

The local squared loss is

$$\ell_{k,i}(\boldsymbol{\theta}; z) = \frac{1}{2}\left[f_{\boldsymbol{\theta}}(i; \mathcal{C}^{\mathrm{dep}}_{i}) - z\right]^2. \tag{30}$$

Define the target-population measurement risk

$$\widetilde{F}_{\boldsymbol{\pi}}(\boldsymbol{\theta}) = \sum_{k=1}^{K}\sum_{i\in\mathcal{I}} \pi^{\mathrm{tar}}_{k,i} \mathbb{E}\left[\ell_{k,i}\left(\boldsymbol{\theta}; Z^{\star}_{k,i}\right)\right]. \tag{31}$$

If

$$\left\|\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(i; \mathcal{C}^{\mathrm{dep}}_{i})\right\|_2 \le G_f, \tag{32}$$

then

$$\left\|\nabla \widetilde{F}_{\boldsymbol{\pi}}(\boldsymbol{\theta}) - \nabla F_{\boldsymbol{\mu}}(\boldsymbol{\theta})\right\|_2 \le G_f \Delta_{\mathrm{cal}}. \tag{33}$$

Thus, exact population calibration makes the target-weighted measurement gradient equal to the public-field target gradient. For real traces, Δ_{cal} explicitly quantifies the remaining population-calibration error.

D. Windowed Asynchronous Federated Learning

The server organizes aggregation decisions into consecutive windows

$$\mathcal{W}_r = [T_r, T_{r+1}), \qquad r = 0, \ldots, R-1. \tag{34}$$

The server keeps model version $\boldsymbol{\theta}_r$ fixed during $\mathcal{W}_r$ and applies one aggregated update when the window closes.

Clients operate asynchronously. An update considered in window r may be trained from an earlier downloaded model

$$\boldsymbol{\theta}_{s_{k,r}}, \qquad s_{k,r} \le r. \tag{35}$$

Its aggregation-time staleness is

$$\tau_{k,r} = r - s_{k,r}. \tag{36}$$

No global-model version is incremented for an individual arrival inside the same window. All usable arrivals in $\mathcal{W}_r$ are jointly calibrated when the window closes.

E. Identifiable Two-Stage Non-Random Selection

A client opportunity affects the global model only after passing two non-random stages: local observation and usable update generation.

1) Local observation: During window r, client k constructs a pre-outcome risk set

$$\mathcal{R}_{k,r} \subseteq \mathcal{I}. \tag{37}$$

It contains all atomic units for which a sensing opportunity is defined in the current local sensing interval. The risk set is determined before the corresponding measurement outcomes and records both successful and unsuccessful opportunities.

For every $i \in \mathcal{R}_{k,r}$, define

$$O_{k,r,i} = \begin{cases} 1, & \text{if a valid measurement is obtained,} \\ 0, & \text{otherwise.} \end{cases} \tag{38}$$

The observation propensity is

$$p^{\mathrm{obs}}_{k,r,i} = \Pr\left(O_{k,r,i} = 1 \mid \boldsymbol{X}^{\mathrm{obs}}_{k,r,i}, \mathcal{H}^{-}_{k,r}\right), \tag{39}$$

where $\boldsymbol{X}^{\mathrm{obs}}_{k,r,i}$ contains only pre-outcome information, such as reachability, distance, time context, sensing schedule, device state, and historical visitation.

The observed buffer is

$$\mathcal{B}_{k,r} = \left\{i \in \mathcal{R}_{k,r} : O_{k,r,i} = 1\right\}. \tag{40}$$

We assume conditional observation ignorability:

$$\boxed{O_{k,r,i} \perp \nabla \ell_{k,i}\left(\boldsymbol{\theta}; Z^{\star}_{k,i}\right) \;\; \boldsymbol{X}^{\mathrm{obs}}_{k,r,i}, \mathcal{H}^{-}_{k,r}.} \tag{41}$$

Unlike an independence condition involving an already expected gradient, (41) concerns the random potential measurement gradient associated with the current opportunity.

2) Usable update generation: Every client with a nonempty observed buffer is registered as a second-stage update attempt:

$$\mathcal{E}_r = \{k \in \mathcal{K} : |\mathcal{B}_{k,r}| > 0\}. \tag{42}$$

If the platform deliberately subsamples these clients, the known attempt-selection probability is included in the second-stage propensity.

For $k \in \mathcal{E}_r$, define

$$U_{k,r} = \mathbf{1}\left\{\begin{array}{l}\text{local training is completed,} \\ \text{the update arrives before } T_{r+1}, \text{ and} \\ \tau_{k,r} \le S_{\max}\end{array}\right\}. \tag{43}$$

Hence, local computation failure, transmission failure, deadline violation, and excessive staleness are represented by a single second-stage outcome.

The usable-update propensity is

$$q^{\mathrm{use}}_{k,r} = \Pr\left(U_{k,r} = 1 \mid \boldsymbol{X}^{\mathrm{use}}_{k,r}, \mathcal{H}^-_r\right). \tag{44}$$

The feature vector $\boldsymbol{X}^{\mathrm{use}}_{k,r}$ may contain pre-training local summaries, expected computation time, available bandwidth, energy, expected transmission delay, downloaded model version, and historical availability. It excludes the current model update, realized computation time, realized transmission delay, and all post-deadline information.

The usable set at the end of the window is

$$\mathcal{A}_r = \{k \in \mathcal{E}_r : U_{k,r} = 1\}. \tag{45}$$

Let

$$\boldsymbol{\Xi}^\star_{k,r} = \left(\boldsymbol{u}^\star_{k,r}, \boldsymbol{c}^\star_{k,r}, v^\star_{k,r}\right) \tag{46}$$

denote the potential model update, group composition, and variance summary that would be produced under successful local computation.

We assume conditional usable-update ignorability:

$$\boxed{U_{k,r} \perp \boldsymbol{\Xi}^\star_{k,r} \; \boldsymbol{X}^{\mathrm{use}}_{k,r}, \mathcal{H}^-_r.} \tag{47}$$

The target-support conditions are

$$\begin{aligned} \pi^{\mathrm{tar}}_{k,s} > 0 \quad \Longrightarrow \quad \pi^{\mathrm{opp}}_{k,s,r} &> 0, \\ p^{\mathrm{obs}}_{k,r,i} &\ge p_0 > 0, \\ q^{\mathrm{use}}_{k,r} &\ge q_0 > 0 \end{aligned} \tag{48}$$

for opportunities used to represent the target support. Equivalently, the ideal design and inverse-propensity weights are required to have bounded moments.

RAVEN-MCS therefore addresses observable, conditionally ignorable selection. It does not identify arbitrary hidden confounding or target regions that are absent from the opportunity process.

F. Target-Risk Misalignment

Let $c^{\mathrm{raw}}_{k,r,i}$ be the normalized raw local weight assigned to atomic unit i by a conventional method, with

$$\sum_{i \in \mathcal{R}_{k,r}} c^{\mathrm{raw}}_{k,r,i} = 1 \tag{49}$$

whenever the client has a nonempty local sample.

Suppose a conventional asynchronous method assigns aggregation weight $\alpha^{\mathrm{base}}_{k,r}$ to usable client k. Conditioned on an active aggregation window, its induced client–unit training distribution is

$$\rho^{\mathrm{base}}_{k,i,r} = \mathbb{E}\left[\alpha^{\mathrm{base}}_{k,r} c^{\mathrm{raw}}_{k,r,i} \,\middle|\, \mathcal{H}^-_r, |\mathcal{A}_r| > 0\right]. \tag{50}$$

It satisfies

$$\sum_{k=1}^{K} \sum_{i \in \mathcal{I}} \rho^{\mathrm{base}}_{k,i,r} = 1. \tag{51}$$

Ignoring finite-sample, local-drift, and stale-model errors, the expected baseline direction is

$$\sum_{k=1}^{K} \sum_{i \in \mathcal{I}} \rho^{\mathrm{base}}_{k,i,r} \boldsymbol{g}_{k,i}(\boldsymbol{\theta}_r), \tag{52}$$

where

$$\boldsymbol{g}_{k,i}(\boldsymbol{\theta}) = \mathbb{E}\left[\nabla \ell_{k,i}\left(\boldsymbol{\theta}; Z^\star_{k,i}\right)\right]. \tag{53}$$

The desired target-population measurement direction is

$$\sum_{k=1}^{K} \sum_{i \in \mathcal{I}} \pi^{\mathrm{tar}}_{k,i} \boldsymbol{g}_{k,i}(\boldsymbol{\theta}_r). \tag{54}$$

In general,

$$\boldsymbol{\rho}^{\mathrm{base}}_r \ne \boldsymbol{\pi}^{\mathrm{tar}}. \tag{55}$$

The discrepancy in (55) may arise from:

1) unequal generation of client–stratum opportunities;
2) unequal atomic-unit opportunities within a stratum;
3) non-random conversion from opportunities to valid measurements;
4) non-random local computation and update delivery; and
5) distorted client composition among measurements associated with the same public target.

We refer to the resulting difference between the induced training direction and the prescribed deployment direction as *target-risk misalignment*.

G. Ideal Two-Stage Corrected Quantities

To formulate the ideal offline problem, suppose that the true design ratios and both selection propensities were known. For an observed opportunity, define the ideal first-stage weight

$$a^\circ_{k,r,i} = \frac{\zeta^\circ_{k,r,i}}{p^{\mathrm{obs}}_{k,r,i}}. \tag{56}$$

The ideal corrected local mass is

$$m^\circ_{k,r} = \sum_{i \in \mathcal{R}_{k,r}} O_{k,r,i} a^\circ_{k,r,i}. \tag{57}$$

For $m^\circ_{k,r} > 0$, the ideal group composition is

$$c^\circ_{k,r,g} = \frac{\displaystyle\sum_{i \in \mathcal{R}_{k,r}: h(i) = g} O_{k,r,i} a^\circ_{k,r,i}}{m^\circ_{k,r}}. \tag{58}$$

Using the true usable-update propensity, the ideal self-normalized second-stage reference is

$$\beta^\circ_{k,r} = \frac{U_{k,r} m^\circ_{k,r} / q^{\mathrm{use}}_{k,r}}{\displaystyle\sum_{j \in \mathcal{E}_r} U_{j,r} m^\circ_{j,r} / q^{\mathrm{use}}_{j,r}}, \qquad k \in \mathcal{A}_r. \tag{59}$$

Let

$$\boldsymbol{M}_r^\circ = \left[\boldsymbol{c}_{k,r}^\circ\right]_{k\in\mathcal{A}_r} \in \mathbb{R}^{G\times K_r}, \qquad K_r = |\mathcal{A}_r|. \tag{60}$$

For aggregation vector $\boldsymbol{\alpha}_r$, the ideal effective group mixture is

$$\boldsymbol{\omega}_r^\circ = \boldsymbol{M}_r^\circ \boldsymbol{\alpha}_r. \tag{61}$$

H. Target-Risk-Calibrated Asynchronous Problem

At the end of an active window, the platform applies

$$\boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r - \eta_r \sum_{k\in\mathcal{A}_r} \alpha_{k,r}\boldsymbol{u}_{k,r}, \tag{62}$$

where $\eta_r > 0$ is the server learning rate.

Define the active-window indicator and cumulative learning mass as

$$I_r = \mathbf{1}\left\{|\mathcal{A}_r| > 0\right\}, \qquad S_R = \sum_{r=0}^{R-1} \eta_r I_r. \tag{63}$$

For a horizon satisfying $S_R > 0$, the learning-rate-weighted long-term group mixture is

$$\overline{\boldsymbol{\omega}}_R^\circ = \frac{\sum_{r=0}^{R-1} \eta_r I_r \boldsymbol{\omega}_r^\circ}{S_R}. \tag{64}$$

Let $\boldsymbol{V}_r^\circ \succeq \mathbf{0}$ be the diagonal matrix of conditional update variances, and define normalized staleness

$$\overline{\tau}_{k,r} = \frac{\tau_{k,r}}{\max\{S_{\max}, 1\}}. \tag{65}$$

The feasible aggregation set is

$$\begin{aligned}\mathcal{C}_r = \Bigg\{\boldsymbol{\alpha} \ : \mathbf{1}^\mathsf{T}\boldsymbol{\alpha} = 1, \quad 0 \le \alpha_k \le \overline{\alpha}_r, \\ \|\boldsymbol{\alpha}\|_2^2 \le \frac{1}{\overline{E}_r}\Bigg\},\end{aligned} \tag{66}$$

where

$$\begin{aligned}\overline{\alpha}_r &= \min\left\{1, \max\left\{\alpha_{\max}, \frac{1}{K_r}\right\}\right\}, \\ \overline{E}_r &= \min\left\{E_{\min}, K_r\right\}.\end{aligned} \tag{67}$$

The uniform vector is feasible for every active window.

The ideal target-risk-calibrated asynchronous problem is

$$\begin{aligned}\mathbf{P1}: \quad \min_{\{\boldsymbol{\theta}_r\},\{\boldsymbol{\alpha}_r\}} \quad & \mathbb{E}\left[F_{\boldsymbol{\mu}}(\boldsymbol{\theta}_R)\right] \\ & + \lambda_L \mathbb{E}\left[\|\overline{\boldsymbol{\omega}}_R^\circ - \boldsymbol{\mu}\|_1\right] \\ & + \lambda_g \mathbb{E}\left[\frac{1}{S_R}\sum_{r=0}^{R-1} \eta_r I_r \left\|\boldsymbol{\omega}_r^\circ - \boldsymbol{\mu}\right\|_2^2\right] \\ & + \lambda_\beta \mathbb{E}\left[\frac{1}{S_R}\sum_{r=0}^{R-1} \eta_r I_r \left\|\boldsymbol{\alpha}_r - \boldsymbol{\beta}_r^\circ\right\|_2^2\right] \\ & + \lambda_v \mathbb{E}\left[\frac{1}{S_R}\sum_{r=0}^{R-1} \eta_r I_r \boldsymbol{\alpha}_r^\mathsf{T} \boldsymbol{V}_r^\circ \boldsymbol{\alpha}_r\right] \\ & + \lambda_s \mathbb{E}\left[\frac{1}{S_R}\sum_{r=0}^{R-1} \eta_r I_r \overline{\boldsymbol{\tau}}_r^\mathsf{T} \boldsymbol{\alpha}_r\right] \qquad \text{(68a)} \\ \text{s.t.} \quad & \boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r - \eta_r \sum_{k\in\mathcal{A}_r} \alpha_{k,r}\boldsymbol{u}_{k,r}, \quad \forall r : I_r = 1, \qquad \text{(68b)} \\ & \boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r, \quad \forall r : I_r = 0, \qquad \text{(68c)} \\ & \boldsymbol{\alpha}_r \in \mathcal{C}_r, \quad \forall r : I_r = 1. \qquad \text{(68d)}\end{aligned}$$

The first term minimizes the public-field completion risk. The second controls long-term target-group representation. The third limits the instantaneous group-mixture error that directly affects the current target gradient. The fourth preserves the client composition recovered by ideal two-stage correction. The final two terms control update variance and model staleness.

Problem $\mathbf{P1}$ cannot be solved offline because future opportunity sets, observation outcomes, local-computation outcomes, transmission delays, usable-update sets, conditional variances, and true selection propensities are unavailable.

RAVEN-MCS therefore constructs lagged estimates of the opportunity design ratio and both selection propensities, applies a stable local Hájek correction, maintains a learning-rate-weighted target-debt process, and solves a strongly convex online calibration problem at the end of each aggregation window.

IV. The Proposed RAVEN-MCS Framework

A. Framework Overview

RAVEN-MCS is a target-risk-calibrated asynchronous federated completion framework for the opportunity imbalance and two-stage non-random selection process defined in Section III. It does not require a new completion backbone. Instead, it corrects the empirical distribution on which the backbone is trained and controls the distribution represented by each asynchronous global update.

During aggregation window $\mathcal{W}_r$, the server keeps $\boldsymbol{\theta}_r$ fixed. Clients independently construct observation risk sets, correct their local objectives, and train from potentially stale downloaded models $\boldsymbol{\theta}_{s_{k,r}}$. All updates that complete computation, arrive before the window deadline, and satisfy the staleness limit form the usable set $\mathcal{A}_r$. When the window closes, the server jointly calibrates these updates and produces $\boldsymbol{\theta}_{r+1}$.

RAVEN-MCS contains five coupled components:

1) lagged estimation of the opportunity-design ratio;
2) stable Hájek correction of local observation selection;
3) inverse-propensity correction of usable update generation;
4) instantaneous and long-term target-group calibration; and
5) variance- and staleness-controlled asynchronous aggregation.

The correction flow is

$$\widehat{\zeta}_{k,r,i}, \widehat{p}^{\mathrm{obs}}_{k,r,i} \longrightarrow a_{k,r,i} \longrightarrow \widehat{F}^{\mathrm{H}}_{k,r} \longrightarrow \boldsymbol{u}_{k,r}, \boldsymbol{c}_{k,r}, m_{k,r} \longrightarrow \widehat{q}^{\mathrm{use}}_{k,r} \longrightarrow \widehat{\boldsymbol{\beta}}_r \longrightarrow \boldsymbol{\alpha}^{\star}_r \longrightarrow \boldsymbol{\theta}_{r+1}.$$

B. Lagged Opportunity-Design Estimation

The target client–stratum masses $\{\pi^{\mathrm{tar}}_{k,s}\}$ and within-stratum target-unit distributions $\{\nu^{\mathrm{tar}}_{i|s}\}$ are fixed before federated model training. The platform estimates only the opportunity process appearing in the denominator of (21).

Let $\mathcal{L}^{\mathrm{opp}}_{r^-}$ be the opportunity log completed before aggregation window r. It contains client identity, opportunity stratum, admissible pre-outcome features, and the atomic unit associated with every recorded opportunity, regardless of whether a valid measurement was eventually obtained.

Using this lagged log, the platform estimates

$$\widehat{\pi}^{\mathrm{opp}}_{k,s,r} = \widehat{\Pr}\left(k, s \mid \mathcal{L}^{\mathrm{opp}}_{r^-}\right) \tag{69}$$

and, when within-stratum exchangeability is not imposed,

$$\widehat{\nu}^{\mathrm{opp}}_{k,r}(i \mid s) = \widehat{\Pr}\left(i \mid k, s, \mathcal{L}^{\mathrm{opp}}_{r^-}\right). \tag{70}$$

These probabilities may be obtained from a known sensing schedule, smoothed stratum counts, or a parametric opportunity model over recurring spatial and temporal features. They are not estimated from one count for each absolute spatio-temporal unit.

The practical design ratio is

$$\boxed{\widehat{\zeta}_{k,r,i} = \frac{\pi^{\mathrm{tar}}_{k,s(i)}}{\max\left\{\widehat{\pi}^{\mathrm{opp}}_{k,s(i),r}, \pi_{\min}\right\}} \frac{\nu^{\mathrm{tar}}_{i|s(i)}}{\max\left\{\widehat{\nu}^{\mathrm{opp}}_{k,r}(i \mid s(i)), \nu_{\min}\right\}},} \tag{71}$$

where $\pi_{\min} > 0$ and $\nu_{\min} > 0$ are numerical floors.

Under the exchangeability condition (22), the second factor equals one, and the default implementation uses

$$\widehat{\zeta}_{k,r,s} = \frac{\pi^{\mathrm{tar}}_{k,s}}{\max\left\{\widehat{\pi}^{\mathrm{opp}}_{k,s,r}, \pi_{\min}\right\}}. \tag{72}$$

All opportunity estimates used in window r are measurable before the current observation outcomes. This prevents the current sensing values or model errors from affecting their own target-design weights.

C. Stable Local Observation Correction

1) Lagged observation-propensity estimation: Client k maintains a local observation-propensity model h^{obs}_k. For every candidate opportunity $i \in \mathcal{R}_{k,r}$, it computes

$$\widehat{p}^{\mathrm{obs}}_{k,r,i} = h^{\mathrm{obs}}_k\left(\boldsymbol{X}^{\mathrm{obs}}_{k,r,i}; \boldsymbol{\phi}^{\mathrm{obs}}_{k,r-1}\right), \tag{73}$$

where $\boldsymbol{\phi}^{\mathrm{obs}}_{k,r-1}$ is trained only on risk-set outcomes completed before window r.

The current observation outcome, potential measurement, and current model error are excluded from their own propensity prediction. The practical combined design-and-observation weight is

$$\boxed{a_{k,r,i} = \min\left\{a_{\max}, \frac{\widehat{\zeta}_{k,r,i}}{\max\left\{\widehat{p}^{\mathrm{obs}}_{k,r,i}, p_{\min}\right\}}\right\}.} \tag{74}$$

The floors control numerical instability, while $a_{\max}$ limits the influence of extremely rare opportunities.

2) Corrected local mass and group composition: The stabilized corrected mass contributed to target group g is

$$m_{k,r,g} = \sum_{i \in \mathcal{R}_{k,r}} O_{k,r,i} a_{k,r,i} \mathbf{1}\left\{h(i) = g\right\}. \tag{75}$$

The total corrected mass is

$$m_{k,r} = \sum_{g=1}^{G} m_{k,r,g} = \sum_{i \in \mathcal{R}_{k,r}} O_{k,r,i} a_{k,r,i}. \tag{76}$$

A client with $m_{k,r} = 0$ does not enter the update-attempt set.

For $m_{k,r} > 0$, define the normalized Hájek weight

$$\bar{a}_{k,r,i} = \frac{O_{k,r,i} a_{k,r,i}}{m_{k,r}}. \tag{77}$$

The practical corrected group composition is

$$c_{k,r,g} = \frac{m_{k,r,g}}{m_{k,r}}, \qquad \sum_{g=1}^{G} c_{k,r,g} = 1. \tag{78}$$

The corresponding effective sample size is

$$n^{\mathrm{eff}}_{k,r} = \frac{m^2_{k,r}}{\sum_{i \in \mathcal{R}_{k,r}} \left(O_{k,r,i} a_{k,r,i}\right)^2} = \frac{1}{\sum_{i \in \mathcal{R}_{k,r}} \bar{a}^2_{k,r,i}}. \tag{79}$$

The two quantities have different meanings. The corrected mass $m_{k,r}$ estimates target-equivalent local opportunity mass and is used in second-stage correction. The effective sample size $n^{\mathrm{eff}}_{k,r}$ measures weight concentration and is used only for uncertainty and stability control.

D. Hájek-Corrected Local Completion

For every successfully observed opportunity, let

$$Z_{k,r,i} = Z^{\star}_{k,i}, \qquad i \in \mathcal{B}_{k,r}. \tag{80}$$

Client k minimizes the self-normalized empirical objective

$$\widehat{F}^{\mathrm{H}}_{k,r}(\boldsymbol{\theta}) = \sum_{i \in \mathcal{B}_{k,r}} \bar{a}_{k,r,i} \ell_{k,i}\left(\boldsymbol{\theta}; Z_{k,r,i}\right). \tag{81}$$

No unobserved potential measurement is required to evaluate (81).

The theoretical algorithm uses local stochastic gradient descent. Starting from $\boldsymbol{\theta}_{s_{k,r}}$, client k performs E_{loc} local steps:

$$\begin{aligned} \boldsymbol{\theta}^{(0)}_{k,r} &= \boldsymbol{\theta}_{s_{k,r}}, \\ \boldsymbol{\theta}^{(e+1)}_{k,r} &= \boldsymbol{\theta}^{(e)}_{k,r} - \gamma_r \widehat{\boldsymbol{g}}^{(e)}_{k,r}, \quad e = 0, \ldots, E_{\mathrm{loc}} - 1, \end{aligned} \tag{82}$$

where $\widehat{\boldsymbol{g}}_{k,r}^{(e)}$ is an unbiased stochastic gradient of (81) conditional on the corrected local sample.

The normalized local displacement is

$$\boldsymbol{u}_{k,r} = \frac{\boldsymbol{\theta}_{s_{k,r}} - \boldsymbol{\theta}_{k,r}^{(E_{\mathrm{loc}})}}{\gamma_r E_{\mathrm{loc}}}. \tag{83}$$

For local SGD, $\boldsymbol{u}_{k,r}$ is the average of the stochastic gradients applied during the local procedure.

E. Attempt Registration and Lagged Variance State

A client with $m_{k,r} > 0$ sends a lightweight control-plane registration before local training:

$$\mathsf{Reg}_{k,r} = \Big(k,\, \boldsymbol{X}_{k,r}^{\mathrm{use}},\, m_{k,r},\, n_{k,r}^{\mathrm{eff}},\, s_{k,r}\Big). \tag{84}$$

The registration message is substantially smaller than a model update. It allows the server to construct the full attempt set $\mathcal{E}_r$ and record a usable-update outcome even if subsequent local computation or model transmission fails.

RAVEN-MCS uses a lagged update-variance state so that the current aggregation decision does not depend on the stochastic noise of the same update. Let $\overline{S}_{k,r^-}^{2}$ be the most recent weighted single-record gradient-variance estimate available before the current attempt. The current variance proxy is

$$v_{k,r} = \frac{\overline{S}_{k,r^-}^{2}}{\max\left\{n_{k,r}^{\mathrm{eff}}, 1\right\}} + v_{\mathrm{floor}}, \tag{85}$$

where $v_{\mathrm{floor}} > 0$ prevents a client without sufficient history from appearing variance-free.

After the local procedure is resolved, the client may update its variance state for future windows. Let $\boldsymbol{g}_{k,r,i}$ be a per-record or micro-batch gradient and define

$$\overline{\boldsymbol{g}}_{k,r} = \sum_{i \in \mathcal{B}_{k,r}} \overline{a}_{k,r,i} \boldsymbol{g}_{k,r,i}. \tag{86}$$

For $n_{k,r}^{\mathrm{eff}} > 1$, the weighted variance estimate is

$$\widehat{S}_{k,r}^{2} = \frac{\displaystyle\sum_{i \in \mathcal{B}_{k,r}} \overline{a}_{k,r,i} \left\|\boldsymbol{g}_{k,r,i} - \overline{\boldsymbol{g}}_{k,r}\right\|_2^2}{1 - \displaystyle\sum_{i \in \mathcal{B}_{k,r}} \overline{a}_{k,r,i}^{2}}. \tag{87}$$

The lagged state is updated as

$$\overline{S}_{k,(r+1)^-}^{2} = \rho_v \overline{S}_{k,r^-}^{2} + (1-\rho_v)\widehat{S}_{k,r}^{2}, \tag{88}$$

where $\rho_v \in [0,1)$. When $n_{k,r}^{\mathrm{eff}} \leq 1$, the previous variance state is retained.

F. Usable-Update Propensity Correction

Upon receiving $\mathsf{Reg}_{k,r}$, the server predicts the usable-update probability using a lagged model:

$$\widehat{q}_{k,r}^{\mathrm{use}} = h^{\mathrm{use}}\left(\boldsymbol{X}_{k,r}^{\mathrm{use}}, \mathcal{H}_r^{-}; \boldsymbol{\phi}_{r-1}^{\mathrm{use}}\right). \tag{89}$$

The estimator uses only information available before the completion and delivery outcomes of the current attempt. It excludes the current update value, realized computation time, realized transmission delay, and future server states.

If client k completes training and its update reaches the server before T_{r+1} with $\tau_{k,r} \leq S_{\max}$, it transmits

$$\mathsf{Upd}_{k,r} = \left(\boldsymbol{u}_{k,r}, \boldsymbol{c}_{k,r}, v_{k,r}\right). \tag{90}$$

Otherwise, its registered outcome is $U_{k,r} = 0$.

For every usable client $k \in \mathcal{A}_r$, define the stabilized inverse usable-update weight

$$d_{k,r} = \min\left\{d_{\max}, \frac{1}{\max\left\{\widehat{q}_{k,r}^{\mathrm{use}}, q_{\min}\right\}}\right\}. \tag{91}$$

The practical two-stage corrected mass is

$$b_{k,r} = m_{k,r} d_{k,r}, \tag{92}$$

and the normalized usable-update reference is

$$\boxed{\widehat{\beta}_{k,r} = \frac{b_{k,r}}{\displaystyle\sum_{j \in \mathcal{A}_r} b_{j,r}}.} \tag{93}$$

The reference client–record mass is

$$\widehat{\Pi}_{k,i,r}^{\mathrm{ref}} = \widehat{\beta}_{k,r} \overline{a}_{k,r,i}, \tag{94}$$

and its client–group marginal is

$$\widehat{\Pi}_{k,g,r}^{\mathrm{ref}} = \widehat{\beta}_{k,r} c_{k,r,g}. \tag{95}$$

When the opportunity model and both selection propensity models are correctly specified and the weights are not truncated, this self-normalized reference estimates the prescribed client–unit target composition. The final aggregation vector is allowed to deviate from the reference to account for finite-window group availability, variance, and staleness.

G. Learning-Rate-Weighted Target Debt

The server maintains a nonnegative target-group debt vector

$$\boldsymbol{Q}_r = [Q_{1,r}, \ldots, Q_{G,r}]^{\mathsf{T}}, \qquad \boldsymbol{Q}_0 = \boldsymbol{0}. \tag{96}$$

For the usable set $\mathcal{A}_r$, define the practical coverage matrix

$$\boldsymbol{M}_r = [\boldsymbol{c}_{k,r}]_{k \in \mathcal{A}_r} \in \mathbb{R}^{G \times K_r}, \qquad K_r = |\mathcal{A}_r|. \tag{97}$$

Given aggregation weights $\boldsymbol{\alpha}_r$, the effective group mixture is

$$\boldsymbol{\omega}_r = \boldsymbol{M}_r \boldsymbol{\alpha}_r. \tag{98}$$

After an active aggregation window, the debt evolves as

$$\boxed{\boldsymbol{Q}_{r+1} = \left[\boldsymbol{Q}_r + \eta_r\left(\boldsymbol{\mu} - \boldsymbol{\omega}_r\right)\right]_+.} \tag{99}$$

When $\mathcal{A}_r = \varnothing$, both the model and debt remain unchanged.

The learning-rate factor gives the debt the same effective-training unit as the global-model update. A group accumulates positive debt when its realized effective training mass is below its prescribed target mass. The projection prevents temporary over-representation from creating unbounded negative credit.

H. Target-Risk-Calibrated Online Aggregation

Consider the Lyapunov function

$$L(\boldsymbol{Q}_r) = \frac{1}{2}\left\|\boldsymbol{Q}_r\right\|_2^2. \quad (100)$$

Using $[x]_+^2 \le x^2$, the one-window debt drift satisfies

$$\begin{aligned} &L(\boldsymbol{Q}_{r+1}) - L(\boldsymbol{Q}_r) \\ &\le \eta_r \boldsymbol{Q}_r^\mathsf{T}\left(\boldsymbol{\mu} - \boldsymbol{M}_r\boldsymbol{\alpha}_r\right) + \frac{\eta_r^2}{2}\left\|\boldsymbol{\mu} - \boldsymbol{M}_r\boldsymbol{\alpha}_r\right\|_2^2. \end{aligned} \quad (101)$$

After division by η_r, the decision-dependent linear drift term is

$$-\boldsymbol{Q}_r^\mathsf{T}\boldsymbol{M}_r\boldsymbol{\alpha}_r. \quad (102)$$

Define the diagonal variance matrix

$$\boldsymbol{V}_r = \operatorname{diag}\left(v_{k,r} : k \in \mathcal{A}_r\right) \succeq \boldsymbol{0}, \quad (103)$$

and normalized aggregation staleness

$$\overline{\tau}_{k,r} = \frac{\tau_{k,r}}{\max\{S_{\max}, 1\}}. \quad (104)$$

The practical feasible set is

$$\begin{aligned} \mathcal{C}_r = \Bigg\{\boldsymbol{\alpha} \ &: \mathbf{1}^\mathsf{T}\boldsymbol{\alpha} = 1, \quad 0 \le \alpha_k \le \overline{\alpha}_r, \\ &\|\boldsymbol{\alpha}\|_2^2 \le \frac{1}{\overline{E}_r}\Bigg\}, \end{aligned} \quad (105)$$

where

$$\begin{aligned} \overline{\alpha}_r &= \min\left\{1, \max\left\{\alpha_{\max}, \frac{1}{K_r}\right\}\right\}, \\ \overline{E}_r &= \min\left\{E_{\min}, K_r\right\}, \qquad 1 \le E_{\min}. \end{aligned} \quad (106)$$

The uniform aggregation vector is feasible for every active window.

RAVEN-MCS selects the aggregation vector by solving

$$\boxed{\begin{aligned} \mathbf{P2}: \quad \min_{\boldsymbol{\alpha}_r \in \mathcal{C}_r} \quad & -\boldsymbol{Q}_r^\mathsf{T}\boldsymbol{M}_r\boldsymbol{\alpha}_r \\ & + \frac{\lambda_g}{2}\left\|\boldsymbol{M}_r\boldsymbol{\alpha}_r - \boldsymbol{\mu}\right\|_2^2 \\ & + \frac{\lambda_\beta}{2}\left\|\boldsymbol{\alpha}_r - \widehat{\boldsymbol{\beta}}_r\right\|_2^2 \\ & + \frac{\lambda_v}{2}\boldsymbol{\alpha}_r^\mathsf{T}\boldsymbol{V}_r\boldsymbol{\alpha}_r \\ & + \lambda_s\overline{\boldsymbol{\tau}}_r^\mathsf{T}\boldsymbol{\alpha}_r. \end{aligned}} \quad (107)$$

For the debt-drift analysis, we set

$$\lambda_g \ge \eta_{\max}, \qquad \eta_{\max} = \sup_r \eta_r. \quad (108)$$

The instantaneous calibration term then dominates the normalized second-order drift term $\eta_r\|\boldsymbol{\mu} - \boldsymbol{M}_r\boldsymbol{\alpha}_r\|_2^2/2$.

The five objective terms have distinct roles:

- the debt term reduces accumulated long-term under-representation;
- the instantaneous group term controls the current group marginal of the target-gradient error;
- the reference term preserves the corrected client–record composition;
- the variance term suppresses statistically unstable updates; and
- the staleness term limits dependence on old downloaded models.

The reference term directly controls the induced client–record distribution. For

$$\widehat{\Pi}_{k,i,r}(\boldsymbol{\alpha}) = \alpha_k \overline{a}_{k,r,i}, \quad (109)$$

we have

$$\begin{aligned} &\left\|\widehat{\boldsymbol{\Pi}}_r(\boldsymbol{\alpha}) - \widehat{\boldsymbol{\Pi}}_r^{\mathrm{ref}}\right\|_1 \\ &= \sum_{k \in \mathcal{A}_r}\sum_{i \in \mathcal{B}_{k,r}}\left|\alpha_k - \widehat{\beta}_{k,r}\right|\overline{a}_{k,r,i} \\ &= \left\|\boldsymbol{\alpha} - \widehat{\boldsymbol{\beta}}_r\right\|_1 \\ &\le \sqrt{K_r}\left\|\boldsymbol{\alpha} - \widehat{\boldsymbol{\beta}}_r\right\|_2. \end{aligned} \quad (110)$$

Problem $\mathbf{P2}$ is a convex quadratically constrained quadratic program. Its objective Hessian is

$$\boldsymbol{H}_r = \lambda_g \boldsymbol{M}_r^\mathsf{T}\boldsymbol{M}_r + \lambda_\beta \boldsymbol{I} + \lambda_v \boldsymbol{V}_r. \quad (111)$$

If $\lambda_\beta > 0$, then

$$\boldsymbol{H}_r \succeq \lambda_\beta \boldsymbol{I}, \quad (112)$$

so the objective is strongly convex and the optimal aggregation vector is unique.

Importantly, $\mathbf{P2}$ uses corrected masses, compositions, lagged variance states, propensity estimates, and staleness, but not the coordinates of the current stochastic model updates. This separation supports the conditional-centering argument in the theoretical analysis.

I. Global Update and Complete Online Procedure

Let $\boldsymbol{\alpha}_r^\star$ be the unique solution of (107). At the end of an active aggregation window, the server updates

$$\boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r - \eta_r \sum_{k \in \mathcal{A}_r} \alpha_{k,r}^\star \boldsymbol{u}_{k,r} \quad (113)$$

and then updates the target debt using (99).

Only after the window is closed are its outcomes included in future estimators:

$$\begin{aligned} \mathcal{L}_{(r+1)^-}^{\mathrm{opp}} &\leftarrow \mathcal{L}_{r^-}^{\mathrm{opp}} \cup \left\{k, s(i), i : i \in \mathcal{R}_{k,r}\right\}, \\ \boldsymbol{\phi}_{k,r}^{\mathrm{obs}} &\leftarrow \mathsf{UpdateObs}\left(\boldsymbol{\phi}_{k,r-1}^{\mathrm{obs}}, \left\{\boldsymbol{X}_{k,r,i}^{\mathrm{obs}}, O_{k,r,i}\right\}_{i \in \mathcal{R}_{k,r}}\right), \\ \boldsymbol{\phi}_r^{\mathrm{use}} &\leftarrow \mathsf{UpdateUse}\left(\boldsymbol{\phi}_{r-1}^{\mathrm{use}}, \left\{\boldsymbol{X}_{k,r}^{\mathrm{use}}, U_{k,r}\right\}_{k \in \mathcal{E}_r}\right). \end{aligned} \quad (114)$$

Algorithm 1 summarizes the complete method.

J. Computational, Communication, and Privacy Scope

For client k, opportunity-design weighting, observation correction, corrected-mass construction, and group summarization require

$$O\left(|\mathcal{R}_{k,r}|\right) \quad (115)$$

additional scalar operations. The dominant client-side computation remains local completion-model training.

Algorithm 1 RAVEN-MCS

Require: Initial model $\boldsymbol{\theta}_0$, target distributions, opportunity strata, $p_{\min}$, $q_{\min}$, $\pi_{\min}$, $\nu_{\min}$, $a_{\max}$, $d_{\max}$, $E_{\min}$, $\alpha_{\max}$, and $S_{\max}$

Initialize $\boldsymbol{Q}_0 \leftarrow \mathbf{0}$, lagged opportunity models, propensity models, and client variance states
for $r = 0, \ldots, R-1$ **do**
  Open aggregation window $\mathcal{W}_r = [T_r, T_{r+1})$ and keep $\boldsymbol{\theta}_r$ fixed
  Estimate lagged opportunity-design ratios $\widehat{\zeta}_{k,r,i}$
  for all $k \in \mathcal{K}$ **in parallel do**
    Construct pre-outcome risk set $\mathcal{R}_{k,r}$
    Estimate $\widehat{p}^{\mathrm{obs}}_{k,r,i}$ from lagged local history
    Compute $a_{k,r,i}$, $m_{k,r,g}$, $m_{k,r}$, and $n^{\mathrm{eff}}_{k,r}$
    if $m_{k,r} > 0$ **then**
      Send $\mathsf{Reg}_{k,r}$ and enter $\mathcal{E}_r$
      Server estimates $\widehat{q}^{\mathrm{use}}_{k,r}$
      Train asynchronously from $\boldsymbol{\theta}_{s_{k,r}}$ using $\widehat{F}^{\mathrm{H}}_{k,r}$
      Construct $\boldsymbol{u}_{k,r}$, $\boldsymbol{c}_{k,r}$, and lagged $v_{k,r}$
      Attempt delivery before T_{r+1}
    end if
  end for
  Close $\mathcal{W}_r$, resolve $\{U_{k,r} : k \in \mathcal{E}_r\}$, and form $\mathcal{A}_r$
  if $\mathcal{A}_r = \varnothing$ **then**
    $\boldsymbol{\theta}_{r+1} \leftarrow \boldsymbol{\theta}_r$ and $\boldsymbol{Q}_{r+1} \leftarrow \boldsymbol{Q}_r$
  else
    Construct $\widehat{\boldsymbol{\beta}}_r$, $\boldsymbol{M}_r$, $\boldsymbol{V}_r$, and $\overline{\boldsymbol{\tau}}_r$
    Solve $\mathbf{P2}$ for $\boldsymbol{\alpha}^{\star}_r$
    Update the global model by (113)
    Update target debt by (99)
  end if
  Update opportunity, propensity, and variance models using completed window-r records
end for
return $\boldsymbol{\theta}_R$

The registration message contains only scalar and low-dimensional control information. A usable update transmits a model displacement of dimension d, a sparse group-composition vector, and a variance scalar. The usable-update communication cost is

$$O\left(d + \mathrm{nnz}(\boldsymbol{c}_{k,r})\right). \tag{116}$$

Problem $\mathbf{P2}$ contains K_r variables. A projected first-order implementation requires approximately

$$O(JGK_r) \tag{117}$$

operations for J iterations, dominated by products involving $\boldsymbol{M}_r$. The calibration dimension is independent of the completion-model dimension d.

RAVEN-MCS preserves raw-measurement locality but does not by itself provide cryptographic privacy or a new differential-privacy guarantee. The registration and group-composition summaries may reveal coarse participation information. Secure aggregation, summary perturbation, or differentially private model updates may be incorporated as orthogonal mechanisms. Any added zero-mean update noise must be included in the variance proxy used by $\mathbf{P2}$.

V. Theoretical Analysis

This section analyzes the statistical correction, target-debt control, and target-risk convergence properties of RAVEN-MCS. We first quantify the distributional bias of conventional asynchronous aggregation. We then bound the perturbation of the practical two-stage reference, establish long-term group-mixture alignment under window reachability, and derive a target-risk convergence result based on the actual per-window client–record distribution error.

A. Assumptions and Filtration

Recall the expected client–unit measurement gradient

$$\boldsymbol{g}_{k,i}(\boldsymbol{\theta}) = \mathbb{E}\left[\nabla \ell_{k,i}\left(\boldsymbol{\theta}; Z^{\star}_{k,i}\right)\right]. \tag{118}$$

The target-population measurement gradient is

$$\widetilde{\boldsymbol{g}}_{\pi}(\boldsymbol{\theta}) = \sum_{k=1}^{K} \sum_{i \in \mathcal{I}} \pi^{\mathrm{tar}}_{k,i} \boldsymbol{g}_{k,i}(\boldsymbol{\theta}). \tag{119}$$

By (33),

$$\left\|\widetilde{\boldsymbol{g}}_{\pi}(\boldsymbol{\theta}) - \nabla F_{\boldsymbol{\mu}}(\boldsymbol{\theta})\right\|_2 \le G_f \Delta_{\mathrm{cal}}. \tag{120}$$

We use the following assumptions.

Assumption 1 (Smoothness and bounded gradients)**.** *The public target risk $F_{\boldsymbol{\mu}}$ is lower bounded by $F^{\star}_{\boldsymbol{\mu}}$. Every client–unit measurement risk is L_c-smooth, and $F_{\boldsymbol{\mu}}$ is L-smooth. There exists $G_0 > 0$ such that*

$$\left\|\boldsymbol{g}_{k,i}(\boldsymbol{\theta})\right\|_2 \le G_0, \qquad \forall k, i, \boldsymbol{\theta}. \tag{121}$$

Assumption 2 (Correction estimation and positivity)**.** *The ignorability and positivity conditions in* (41), (47)*, and* (48) *hold. The lagged estimators satisfy*

$$\begin{aligned} \max_{k,i}\left|\widehat{\zeta}_{k,r,i} - \zeta^{\circ}_{k,r,i}\right| &\le \epsilon^{\zeta}_r, \\ \max_{k,i}\left|\widehat{p}^{\mathrm{obs}}_{k,r,i} - p^{\mathrm{obs}}_{k,r,i}\right| &\le \epsilon^{p}_r, \\ \max_{k}\left|\widehat{q}^{\mathrm{use}}_{k,r} - q^{\mathrm{use}}_{k,r}\right| &\le \epsilon^{q}_r. \end{aligned} \tag{122}$$

Assumption 3 (Weight regularity)**.** *The ideal design and inverse-propensity weights have bounded second moments. For every active window, the total ideal first-stage and second-stage corrected masses are bounded away from zero. Moreover,*

$$0 < v_{k,r} \le V_{\max} < \infty. \tag{123}$$

Let F^{cal}_r be the sigma-field generated by all information used to solve $\mathbf{P2}$, including opportunity and propensity estimates, observation indicators, corrected masses, group compositions, registered usable-update outcomes, lagged variance states, and staleness values. It excludes the stochastic gradient noise of the current local updates.

Because $\mathbf{P2}$ does not use the coordinates of $\boldsymbol{u}_{k,r}$, the aggregation vector $\boldsymbol{\alpha}_r$ is measurable with respect to F^{cal}_r.

B. Distributional Misalignment of Conventional Aggregation

The following proposition formalizes the motivating failure mode.

Proposition 1 (Target-gradient misalignment). *Let $\boldsymbol{\rho}_r^{\mathrm{base}}$ be the client–unit distribution induced by a conventional asynchronous method as defined in* (50). *Ignoring local-optimization, stochastic, and stale-model errors,*

$$\begin{aligned}&\left\|\mathbb{E}\left[\widehat{\boldsymbol{g}}_r^{\mathrm{base}} \mid \mathcal{H}_r^-, I_r=1\right]-\nabla F_{\boldsymbol{\mu}}(\boldsymbol{\theta}_r)\right\|_2\\&\leq G_0\left\|\boldsymbol{\rho}_r^{\mathrm{base}}-\boldsymbol{\pi}^{\mathrm{tar}}\right\|_1+G_f\Delta_{\mathrm{cal}}.\end{aligned}\tag{124}$$

Proof. The difference between the baseline direction and the target-population measurement direction equals

$$\sum_{k,i}\left(\rho_{k,i,r}^{\mathrm{base}}-\pi_{k,i}^{\mathrm{tar}}\right)\boldsymbol{g}_{k,i}(\boldsymbol{\theta}_r).\tag{125}$$

Its norm is bounded by $G_0\|\boldsymbol{\rho}_r^{\mathrm{base}}-\boldsymbol{\pi}^{\mathrm{tar}}\|_1$. Equation (120) then gives the result. □

Proposition 1 shows that accurate optimization over the received-update distribution need not minimize the prescribed public-field risk.

C. Local Update Decomposition

For a usable client, define the conditional weighted measurement gradient at its downloaded model as

$$\boldsymbol{g}_{k,r}^{\mathrm{H}}=\sum_{i\in\mathcal{B}_{k,r}}\bar{a}_{k,r,i}\boldsymbol{g}_{k,i}\left(\boldsymbol{\theta}_{s_{k,r}}\right).\tag{126}$$

The normalized local displacement can be decomposed as

$$\boldsymbol{u}_{k,r}=\boldsymbol{g}_{k,r}^{\mathrm{H}}+\boldsymbol{e}_{k,r}^{\mathrm{sgd}}+\boldsymbol{\xi}_{k,r},\tag{127}$$

where $\boldsymbol{e}_{k,r}^{\mathrm{sgd}}$ is the deterministic local-optimization drift and $\boldsymbol{\xi}_{k,r}$ is the stochastic local-gradient error.

Assumption 4 (Local SGD and conditional centering). *There exists a constant $C_{\mathrm{sgd}}>0$ such that*

$$\left\|\boldsymbol{e}_{k,r}^{\mathrm{sgd}}\right\|_2\leq C_{\mathrm{sgd}}\gamma_r E_{\mathrm{loc}},\tag{128}$$

and

$$\mathbb{E}\left[\boldsymbol{\xi}_{k,r}\mid F_r^{\mathrm{cal}}\right]=\mathbf{0}.\tag{129}$$

The lagged variance construction is essential for (129). A same-update variance estimate could make $\boldsymbol{\alpha}_r$ a function of the current stochastic update noise.

D. Perturbation of the Two-Stage Reference

For an observed opportunity, define the untruncated ideal weight

$$a_{k,r,i}^{\circ}=\frac{\zeta_{k,r,i}^{\circ}}{p_{k,r,i}^{\mathrm{obs}}}.\tag{130}$$

Define the true-but-stabilized weight

$$\widetilde{a}_{k,r,i}^{\circ}=\min\left\{a_{\max},\frac{\zeta_{k,r,i}^{\circ}}{\max\left\{p_{k,r,i}^{\mathrm{obs}},p_{\min}\right\}}\right\}.\tag{131}$$

The first-stage stabilization bias is summarized by

$$B_{k,r}^{(1)}=\frac{\sum_{i\in\mathcal{R}_{k,r}}O_{k,r,i}\left|a_{k,r,i}^{\circ}-\widetilde{a}_{k,r,i}^{\circ}\right|}{\max\left\{m_{k,r}^{\circ},m_{\min}\right\}}.\tag{132}$$

For the second stage, define

$$d_{k,r}^{\circ}=\frac{1}{q_{k,r}^{\mathrm{use}}}\tag{133}$$

and

$$\widetilde{d}_{k,r}^{\circ}=\min\left\{d_{\max},\frac{1}{\max\left\{q_{k,r}^{\mathrm{use}},q_{\min}\right\}}\right\}.\tag{134}$$

The corresponding second-stage stabilization bias is

$$B_r^{(2)}=\frac{\sum_{k\in\mathcal{A}_r}m_{k,r}^{\circ}\left|d_{k,r}^{\circ}-\widetilde{d}_{k,r}^{\circ}\right|}{\max\left\{\sum_{k\in\mathcal{A}_r}m_{k,r}^{\circ},m_{\min}\right\}}.\tag{135}$$

Let $\bar{a}_{k,r,i}^{\circ}$ be the ideal normalized first-stage weight, and define the ideal finite-window client–record reference

$$\Pi_{k,i,r}^{\circ}=\beta_{k,r}^{\circ}\bar{a}_{k,r,i}^{\circ}.\tag{136}$$

Its finite-window opportunity error is

$$\epsilon_r^{\mathrm{opp}}=\left\|\boldsymbol{\Pi}_r^{\circ}-\boldsymbol{\pi}^{\mathrm{tar}}\right\|_1.\tag{137}$$

The practical reference distribution is

$$\widehat{\Pi}_{k,i,r}^{\mathrm{ref}}=\widehat{\beta}_{k,r}\bar{a}_{k,r,i}.\tag{138}$$

Lemma 1 (Two-stage reference error). *Under Assumptions 2 and 3, there exist finite constants $C_\zeta,C_p,C_q,C_1,C_2>0$, independent of R, such that*

$$\begin{aligned}&\left\|\widehat{\boldsymbol{\Pi}}_r^{\mathrm{ref}}-\boldsymbol{\pi}^{\mathrm{tar}}\right\|_1\\&\leq\epsilon_r^{\mathrm{ref}}\triangleq\epsilon_r^{\mathrm{opp}}+C_\zeta\epsilon_r^{\zeta}+C_p\epsilon_r^{p}+C_q\epsilon_r^{q}+C_1B_r^{(1)}+C_2B_r^{(2)}.\end{aligned}\tag{139}$$

The term $\epsilon_r^{\mathrm{opp}}$ remains even with oracle design ratios and propensities because one finite aggregation window need not contain a perfectly representative realization of the target opportunity population.

E. Calibration Stability and Pair-Distribution Error

For aggregation vector $\boldsymbol{\alpha}$, define its induced client–record distribution as

$$\widehat{\Pi}_{k,i,r}(\boldsymbol{\alpha})=\alpha_k\bar{a}_{k,r,i},\qquad k\in\mathcal{A}_r,\quad i\in\mathcal{B}_{k,r}.\tag{140}$$

It is normalized because

$$\sum_{k,i}\widehat{\Pi}_{k,i,r}(\boldsymbol{\alpha})=\sum_k\alpha_k=1.\tag{141}$$

The per-window client–record discrepancy is

$$\boxed{\delta_r^{\mathrm{pair}}=\left\|\widehat{\boldsymbol{\Pi}}_r(\boldsymbol{\alpha}_r)-\boldsymbol{\pi}^{\mathrm{tar}}\right\|_1.}\tag{142}$$

Its group marginal discrepancy is

$$\delta_r^{\rm grp} = \left\| \boldsymbol{M}_r \boldsymbol{\alpha}_r - \boldsymbol{\mu} \right\|_1 . \tag{143}$$

Marginalization cannot increase total variation, so

$$\delta_r^{\rm grp} \le \delta_r^{\rm pair}. \tag{144}$$

Using (110) and Lemma 1,

$$\boxed{\delta_r^{\rm pair} \le \sqrt{K_r} \left\| \boldsymbol{\alpha}_r - \widehat{\boldsymbol{\beta}}_r \right\|_2 + \epsilon_r^{\rm ref}.} \tag{145}$$

For fixed $\boldsymbol{Q}_r$, $\boldsymbol{M}_r$, $\boldsymbol{V}_r$, and $\overline{\boldsymbol{\tau}}_r$, let $\mathcal{A}_r(\boldsymbol{\beta})$ denote the optimizer of $\mathbf{P2}$ with reference vector $\boldsymbol{\beta}$.

Proposition 2 (Reference non-expansiveness)**.** *For any two reference vectors $\boldsymbol{\beta}$ and $\boldsymbol{\beta}'$,*

$$\left\| \mathcal{A}_r(\boldsymbol{\beta}) - \mathcal{A}_r(\boldsymbol{\beta}') \right\|_2 \le \left\| \boldsymbol{\beta} - \boldsymbol{\beta}' \right\|_2 . \tag{146}$$

Proof. Collect all terms in $\mathbf{P2}$ except the reference penalty into a closed proper convex function Ψ_r that includes the indicator of $\mathcal{C}_r$. The optimizer can be written as

$$\mathcal{A}_r(\boldsymbol{\beta}) = \mathrm{prox}_{\Psi_r/\lambda_\beta}(\boldsymbol{\beta}). \tag{147}$$

The result follows from non-expansiveness of the proximal mapping. □

F. Target Debt and Long-Term Group Alignment

We first give an exact deterministic relation.

Theorem 1 (Debt-to-mixture relation)**.** *For every horizon satisfying $S_R > 0$,*

$$\left\| \overline{\boldsymbol{\omega}}_R - \boldsymbol{\mu} \right\|_1 \le \frac{2\|\boldsymbol{Q}_R\|_1}{S_R}, \tag{148}$$

where

$$\overline{\boldsymbol{\omega}}_R = \frac{\sum_{r=0}^{R-1} \eta_r I_r \boldsymbol{\omega}_r}{S_R}. \tag{149}$$

Proof. Define

$$\boldsymbol{D}_R = \sum_{r=0}^{R-1} \eta_r I_r \left(\boldsymbol{\mu} - \boldsymbol{\omega}_r \right). \tag{150}$$

Since $[x]_+ \ge x$, the debt recursion implies $D_{R,g} \le Q_{g,R}$. Moreover, $\mathbf{1}^\mathsf{T} \boldsymbol{D}_R = 0$ because both $\boldsymbol{\mu}$ and $\boldsymbol{\omega}_r$ are probability vectors. Therefore,

$$\|\boldsymbol{D}_R\|_1 = 2 \sum_g [D_{R,g}]_+ \le 2\|\boldsymbol{Q}_R\|_1. \tag{151}$$

Using $\boldsymbol{D}_R = S_R(\boldsymbol{\mu} - \overline{\boldsymbol{\omega}}_R)$ completes the proof. □

Theorem 1 is conditional on the target debt. To control debt growth, the target must be reachable over bounded windows.

Assumption 5 (Approximate window reachability)**.** *There exist an integer $W < \infty$ and $\epsilon_{\rm reach} \ge 0$ such that every block $\mathcal{B}$ containing at most W consecutive active aggregation windows admits feasible comparator vectors $\widetilde{\boldsymbol{\alpha}}_r \in \mathcal{C}_r$ satisfying*

$$\begin{aligned} &\left\| \sum_{r \in \mathcal{B}} \eta_r \left(\boldsymbol{M}_r \widetilde{\boldsymbol{\alpha}}_r - \boldsymbol{\mu} \right) \right\|_2 \\ &\qquad \le \epsilon_{\rm reach} \sum_{r \in \mathcal{B}} \eta_r. \end{aligned} \tag{152}$$

Exact reachability corresponds to $\epsilon_{\rm reach} = 0$. The target need not be feasible in every individual aggregation window.

Theorem 2 (Target-debt growth)**.** *Suppose Assumptions 3 and 5 hold and*

$$0 < \eta_r \le \eta_{\max}. \tag{153}$$

Then there exist constants $C_Q, C_{\rm reach}, C_0 > 0$, independent of R, such that

$$\begin{aligned} \|\boldsymbol{Q}_R\|_2 \le & C_Q \sqrt{C_0 + S_R + \sum_{r=0}^{R-1} \eta_r^2 I_r} \\ & + C_{\rm reach} \epsilon_{\rm reach} S_R. \end{aligned} \tag{154}$$

Consequently,

$$\begin{aligned} &\left\| \overline{\boldsymbol{\omega}}_R - \boldsymbol{\mu} \right\|_1 \\ &\le \frac{2\sqrt{G} C_Q}{S_R} \sqrt{C_0 + S_R + \sum_r \eta_r^2 I_r} + 2\sqrt{G} C_{\rm reach} \epsilon_{\rm reach}. \end{aligned} \tag{155}$$

Under exact window reachability and bounded server learning rates, the long-term group mismatch is $O(S_R^{-1/2})$. Approximate reachability produces a residual proportional to $\epsilon_{\rm reach}$.

The constants depend on the maximum aggregation-window length, bounded non-debt penalties in $\mathbf{P2}$, the group dimension, and the maximum server step, but not on the training horizon.

G. Aggregate Update Variance

Assumption 6 (Variance-proxy calibration)**.** *Conditional on $F_r^{\rm cal}$, the stochastic errors of different usable clients are uncorrelated, and*

$$\mathbb{E} \left[\|\boldsymbol{\xi}_{k,r}\|_2^2 \mid F_r^{\rm cal} \right] \le c_v v_{k,r} + \epsilon_v \tag{156}$$

for constants $c_v, \epsilon_v \ge 0$.

Lemma 2 (Aggregate variance bound)**.** *Define*

$$\Sigma_r^2 = \mathbb{E} \left[\left\| \sum_{k \in \mathcal{A}_r} \alpha_{k,r} \boldsymbol{\xi}_{k,r} \right\|_2^2 \middle| F_r^{\rm cal} \right]. \tag{157}$$

Then

$$\Sigma_r^2 \le c_v \boldsymbol{\alpha}_r^\mathsf{T} \boldsymbol{V}_r \boldsymbol{\alpha}_r + \frac{\epsilon_v}{\overline{E}_r}. \tag{158}$$

Proof. Conditional uncorrelatedness gives

$$\Sigma_r^2 = \sum_{k \in \mathcal{A}_r} \alpha_{k,r}^2 \mathbb{E} \left[\|\boldsymbol{\xi}_{k,r}\|_2^2 \mid F_r^{\rm cal} \right]. \tag{159}$$

Applying (156) and $\|\boldsymbol{\alpha}_r\|_2^2 \le 1/\overline{E}_r$ yields the result. □

H. Per-Window Target-Gradient Bias

Define the aggregate update direction

$$\widehat{\boldsymbol{g}}_r = \sum_{k \in \mathcal{A}_r} \alpha_{k,r} \boldsymbol{u}_{k,r}. \tag{160}$$

The aggregate local-SGD drift is

$$\Delta_r^{\rm sgd} = \sum_{k \in \mathcal{A}_r} \alpha_{k,r} \left\| \boldsymbol{e}_{k,r}^{\rm sgd} \right\|_2 . \tag{161}$$

By (128),

$$\Delta_r^{\mathrm{sgd}} \le C_{\mathrm{sgd}} \gamma_r E_{\mathrm{loc}}. \tag{162}$$

The stale-model discrepancy is

$$\Delta_r^{\mathrm{stale}} = L_c U \sum_{k \in \mathcal{A}_r} \alpha_{k,r} \sum_{t=s_{k,r}}^{r-1} \eta_t I_t, \tag{163}$$

where the bounded model-movement condition is

$$\|\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t\|_2 \le \eta_t U. \tag{164}$$

Lemma 3 (Per-window target-gradient bias)**.** *Under Assumptions 1 and 4,*

$$\begin{aligned} &\left\|\mathbb{E}\left[\widehat{\boldsymbol{g}}_r \mid F_r^{\mathrm{cal}}\right] - \nabla F_{\boldsymbol{\mu}}(\boldsymbol{\theta}_r)\right\|_2 \\ &\le B_r \triangleq G_0 \delta_r^{\mathrm{pair}} + \Delta_r^{\mathrm{sgd}} + \Delta_r^{\mathrm{stale}} + G_f \Delta_{\mathrm{cal}}. \end{aligned} \tag{165}$$

Proof. Using (127) and conditional centering, the conditional mean aggregate direction is the $\widehat{\boldsymbol{\Pi}}_r(\boldsymbol{\alpha}_r)$-weighted measurement gradient evaluated at the downloaded client models, plus local-SGD drift.

Replacing the induced client–record distribution by $\boldsymbol{\pi}^{\mathrm{tar}}$ contributes at most $G_0 \delta_r^{\mathrm{pair}}$. Replacing each downloaded model by $\boldsymbol{\theta}_r$ contributes at most $\Delta_r^{\mathrm{stale}}$. The local-SGD approximation contributes Δ_r^{sgd}. Finally, (120) contributes $G_f \Delta_{\mathrm{cal}}$. □

Combining (145) and (165) gives the observable upper bound

$$\begin{aligned} B_r \le & G_0 \sqrt{K_r} \left\|\boldsymbol{\alpha}_r - \widehat{\boldsymbol{\beta}}_r\right\|_2 + G_0 \epsilon_r^{\mathrm{ref}} \\ & + \Delta_r^{\mathrm{sgd}} + \Delta_r^{\mathrm{stale}} + G_f \Delta_{\mathrm{cal}}. \end{aligned}$$

The instantaneous group-calibration term in $\mathbf{P2}$ separately controls the group marginal δ_r^{grp}, whereas the reference penalty controls the finer client–record composition appearing in (166).

I. Target-Risk Convergence

The target-risk result is based on the actual per-window gradient bias in Lemma 3. Long-term group-mixture alignment is not used as a substitute for per-window target-gradient control.

Theorem 3 (Target-risk convergence)**.** *Suppose Assumptions 1, 4, and 6 hold, and*

$$0 < \eta_r \le \overline{\eta} < \frac{1}{4L}. \tag{166}$$

Then there exist constants $C_0, C_1, C_2 > 0$, independent of R, such that

$$\begin{aligned} &\frac{\sum_{r=0}^{R-1} \eta_r I_r \mathbb{E}\left[\|\nabla F_{\boldsymbol{\mu}}(\boldsymbol{\theta}_r)\|_2^2\right]}{S_R} \\ &\le \frac{C_0 \left[F_{\boldsymbol{\mu}}(\boldsymbol{\theta}_0) - F_{\boldsymbol{\mu}}^\star\right]}{S_R} \\ &\quad + C_1 \frac{\sum_{r=0}^{R-1} \eta_r I_r \mathbb{E}\left[B_r^2\right]}{S_R} \\ &\quad + C_2 L \frac{\sum_{r=0}^{R-1} \eta_r^2 I_r \mathbb{E}\left[G_0^2 + B_r^2 + \Sigma_r^2\right]}{S_R}. \end{aligned}$$

The first term in (167) is the finite-horizon optimization error. The second term is the weighted persistent target-gradient bias. The final term contains stochastic update variance and higher-order smoothness effects.

RAVEN-MCS converges to a neighborhood of a stationary point when design-ratio error, propensity error, clipping bias, finite-window opportunity error, population-calibration residual, local-SGD drift, and staleness remain nonzero.

Exact stationary convergence requires

$$S_R \to \infty, \tag{167}$$

$$\frac{\sum_r \eta_r I_r \mathbb{E}[B_r^2]}{S_R} \to 0, \tag{168}$$

and

$$\frac{\sum_r \eta_r^2 I_r \mathbb{E}[\Sigma_r^2]}{S_R} \to 0. \tag{169}$$

The debt and convergence results have different meanings. Theorems 1 and 2 control the long-term target-group mixture. Theorem 3 shows that target-risk optimization additionally depends on the complete per-window client–record discrepancy and on residual measurement, optimization, staleness, and variance errors.

VI. Performance Evaluation

We evaluate RAVEN-MCS through the following research questions.

- **RQ1:** Do opportunity imbalance, non-random observation, and non-random usable-update generation cause conventional asynchronous learning to optimize a distribution different from the prescribed deployment target?
- **RQ2:** Does RAVEN-MCS reduce target-weighted completion error, particularly for target units and groups that receive insufficient raw training mass?
- **RQ3:** Are opportunity-design correction, observation correction, usable-update correction, instantaneous group calibration, long-term target debt, variance control, and staleness control all necessary?
- **RQ4:** How sensitive is RAVEN-MCS to design-ratio estimation error, propensity misspecification, weight truncation, opportunity drift, limited support, and population calibration residual?

TABLE I: Datasets and experimental roles.

Dataset	Signal	Experimental role
Sensor-Scope	Environment	Controlled sensing and completion benchmark
U-Air	$PM_{2.5}$	Spatially heterogeneous urban sensing benchmark
Traffic Volume	Traffic	Benchmark with strong temporal dependence
T-Drive-Speed	Speed	Trace-consistent benchmark with real client mobility

- **RQ5:** Do target debt, window reachability, client–record distribution error, and aggregate variance behave consistently with the theoretical results in Section V?
- **RQ6:** What computation, communication, memory, and convergence-time overhead does RAVEN-MCS introduce?

A. Datasets and Public-Field Construction

We use three controlled completion benchmarks and one trace-consistent mobile crowdsensing benchmark, as summarized in Table I.

For Sensor-Scope, U-Air, and Traffic Volume, preprocessing constructs a public reference field

$$\boldsymbol{Y} = \{Y_i : i \in \mathcal{I}\}. \tag{170}$$

The simulator initially hides all training entries. An entry becomes locally available only when the corresponding client receives a pre-observation opportunity and the observation outcome is positive. Unobserved entries remain unavailable to clients, the platform, the completion model, and all propensity estimators.

For T-Drive-Speed, vehicles are partitioned by identity into a reference fleet and a client fleet. The split is stratified by trajectory length and regenerated for each random seed. The reference fleet is used only to construct the public speed field:

$$Y_{n,t} = \frac{1}{|\mathcal{V}_{n,t}^{\mathrm{ref}}|} \sum_{v \in \mathcal{V}_{n,t}^{\mathrm{ref}}} z_{v,n,t}, \tag{171}$$

where $\mathcal{V}_{n,t}^{\mathrm{ref}}$ is the set of reference vehicles with valid records in cell n during interval t. Only atomic units containing at least $M_{\min}^{\mathrm{ref}}$ reference records are included in the public evaluation support.

Vehicles in the disjoint client fleet act as federated clients and supply local measurements. Thus, no vehicle record is simultaneously used as a client measurement and as its own public target.

Each dataset is divided chronologically into training, validation, and test periods. The early part of the training period is used as a warm-up interval for opportunity estimation, propensity initialization, and variance state initialization. No validation or test completion error is used to estimate opportunity-design ratios or selection propensities.

B. Target Groups and Repeatable Opportunity Strata

Target groups are defined by spatial region and coarse time-of-day block:

$$h(i) = \big(n(i), b_h(t(i))\big). \tag{172}$$

The default setting uses four time-of-day blocks.

Opportunity strata must recur across multiple days and cannot contain an absolute time-slot identifier. We use

$$s(i) = \big(n_s(i), b_s(t(i)), d(i)\big), \tag{173}$$

where $n_s(i)$ is a spatial region, $b_s(t(i))$ is a time-of-day interval, and $d(i)$ distinguishes weekday and weekend opportunities. The stratum resolution is selected using validation data.

The primary public target is uniform across supported atomic units:

$$\varpi_i^{\mathrm{tar}} = \frac{1}{|\mathcal{I}^{\mathrm{sup}}|}, \qquad i \in \mathcal{I}^{\mathrm{sup}}. \tag{174}$$

The corresponding $\boldsymbol{\mu}$, $\nu_{i|g}^{\mathrm{tar}}$, and Λ_s^{tar} are obtained by marginalization.

We additionally evaluate a service-weighted target:

$$\varpi_i^{\mathrm{tar}} = \frac{w_i^{\mathrm{svc}}}{\sum_{\ell \in \mathcal{I}^{\mathrm{sup}}} w_\ell^{\mathrm{svc}}}, \tag{175}$$

where w_i^{svc} is fixed before training from application-side information such as monitoring priority, road category, or administrative importance.

For the controlled benchmarks, $\lambda_{k|s}^{\mathrm{tar}}$ is specified by the population generator. For T-Drive-Speed, it is estimated from warm-up risk-set opportunities:

$$\widehat{\lambda}_{k|s}^{\mathrm{tar}} = \frac{N_{k,s}^{\mathrm{opp}} + \xi_\lambda \lambda_{k|s}^{\mathrm{base}}}{\sum_j \left(N_{j,s}^{\mathrm{opp}} + \xi_\lambda \lambda_{j|s}^{\mathrm{base}}\right)}, \tag{176}$$

where $N_{k,s}^{\mathrm{opp}}$ is the number of warm-up opportunities for client k in recurring stratum s. The smoothing distribution has support only on client–stratum pairs observed during the warm-up period.

C. Opportunity Process and Design-Ratio Estimation

1) Controlled opportunity generator: For the controlled benchmarks, each client follows a stochastic mobility and sensing-schedule process. At aggregation window r, its current mobility state, reachable regions, sensing schedule, and device status determine $\mathcal{R}_{k,r}$.

The client–stratum opportunity probability is generated by

$$\pi_{k,s,r}^{\mathrm{opp}} \propto \exp\left(\boldsymbol{\vartheta}_k^{\mathsf{T}} \boldsymbol{\psi}_{s,r} + \chi_{k,s,r}\right), \tag{177}$$

where $\boldsymbol{\psi}_{s,r}$ contains recurring spatial and temporal features and $\chi_{k,s,r}$ controls client-specific preference and temporal drift.

Within each stratum, atomic units are sampled according to

$$\nu_{k,r}^{\mathrm{opp}}(i \mid s). \tag{178}$$

The primary experiments impose within-stratum exchangeability:

$$\nu_{k,r}^{\mathrm{opp}}(i \mid s) = \nu_{i|s}^{\mathrm{tar}}. \tag{179}$$

A robustness experiment introduces within-stratum imbalance so that both factors in (21) are required.

Temporal opportunity drift is controlled by

$$\Gamma_{\mathrm{drift}} = \frac{1}{R-1} \sum_{r=1}^{R-1} \left\| \boldsymbol{\pi}_{r+1}^{\mathrm{opp}} - \boldsymbol{\pi}_r^{\mathrm{opp}} \right\|_1. \tag{180}$$

2) Practical opportunity estimation: The practical implementation estimates $\pi^{\mathrm{opp}}_{k,s,r}$ from lagged recurring-stratum counts with exponential forgetting:

$$C^{\mathrm{opp}}_{k,s,r} = \rho_{\mathrm{opp}} C^{\mathrm{opp}}_{k,s,r-1} + N^{\mathrm{opp}}_{k,s,r-1}, \tag{181}$$

where $N^{\mathrm{opp}}_{k,s,r-1}$ is the number of opportunities completed in the previous window. The estimated probability is

$$\widehat{\pi}^{\mathrm{opp}}_{k,s,r} = \frac{C^{\mathrm{opp}}_{k,s,r} + \xi_\pi \pi^{\mathrm{base}}_{k,s}}{\sum_{j,u} \left(C^{\mathrm{opp}}_{j,u,r} + \xi_\pi \pi^{\mathrm{base}}_{j,u} \right)}. \tag{182}$$

When within-stratum exchangeability is used, the design ratio becomes

$$\widehat{\zeta}_{k,r,s} = \frac{\pi^{\mathrm{tar}}_{k,s}}{\max\left\{ \widehat{\pi}^{\mathrm{opp}}_{k,s,r}, \pi_{\min} \right\}}. \tag{183}$$

For non-exchangeable settings, a multinomial conditional-unit model estimates $\nu^{\mathrm{opp}}_{k,r}(i \mid s)$.

Controlled experiments include an oracle design-ratio condition. The real-trace experiments use only lagged estimated ratios.

D. Client Measurement Heterogeneity

For the controlled benchmarks, local measurements are generated as

$$Z^\star_{k,i} = Y_i + b_{k,i} + \varepsilon_{k,i}, \tag{184}$$

where $\varepsilon_{k,i}$ is zero-mean random noise.

Uncentered client effects $\widetilde{b}_{k,i}$ are generated according to the prescribed heterogeneity level σ_b. They are then centered under the target client composition:

$$b_{k,i} = \widetilde{b}_{k,i} - \sum_j \lambda^{\mathrm{tar}}_{j|s(i)} \widetilde{b}_{j,i}. \tag{185}$$

Hence,

$$\Delta_{\mathrm{cal}} = 0 \tag{186}$$

in the principal controlled experiments.

A calibration-residual stress test adds a common stratum-dependent offset:

$$b^{\mathrm{stress}}_{k,i} = b_{k,i} + \delta^{\mathrm{cal}}_{s(i)}. \tag{187}$$

For T-Drive-Speed, client heterogeneity is not synthetically centered. We estimate a validation-period population-calibration residual:

$$\widehat{\Delta}_{\mathrm{cal}} = \sum_{s \in \mathcal{S}} \Lambda^{\mathrm{tar}}_s \left| \sum_k \widehat{\lambda}^{\mathrm{tar}}_{k|s} \widehat{b}_{k,s} \right|, \tag{188}$$

where $\widehat{b}_{k,s}$ is the difference between client k's mean measurement and the reference-fleet mean within stratum s. This quantity is reported as a validity diagnostic and is not forced to zero by preprocessing.

E. Two-Stage Selection Processes

1) Local observation selection: For each $i \in \mathcal{R}_{k,r}$, the controlled observation propensity is

$$p^{\mathrm{obs}}_{k,r,i} = \mathrm{clip}\left[\sigma\left(a_0 + \boldsymbol{a}^\top \boldsymbol{X}^{\mathrm{obs}}_{k,r,i} \right), p_{\mathrm{lo}}, p_{\mathrm{hi}} \right]. \tag{189}$$

The covariates include reachability, distance, time context, device state, sensing schedule, and historical visitation. The outcome is sampled as

$$O_{k,r,i} \sim \mathrm{Bernoulli}\left(p^{\mathrm{obs}}_{k,r,i} \right). \tag{190}$$

The current latent value, current measurement error, and current model error do not enter the generator.

For T-Drive-Speed, the trajectory and pre-record metadata determine the risk set and observation features. A measurement is regarded as valid according to timestamp, physical-range, and packet-integrity rules fixed before model training.

2) Windowed usable-update generation: The server uses fixed aggregation windows

$$\mathcal{W}_r = [T_r, T_{r+1}), \qquad T_{r+1} - T_r = \Delta T. \tag{191}$$

The model version remains unchanged inside a window and is incremented only after the joint update at its closure.

For every registered attempt $k \in \mathcal{E}_r$, the simulator generates a local-computation success indicator

$$C^{\mathrm{cmp}}_{k,r} \sim \mathrm{Bernoulli}\left(p^{\mathrm{cmp}}_{k,r} \right) \tag{192}$$

and, conditioned on success, a computation duration

$$T^{\mathrm{cmp}}_{k,r} \sim \mathrm{LogNormal}\left(\mu^{\mathrm{cmp}}_{k,r}, \sigma^2_{\mathrm{cmp}} \right). \tag{193}$$

The parameters depend on device capability, energy, corrected local mass, and local training steps.

Network success is generated as

$$C^{\mathrm{net}}_{k,r} \sim \mathrm{Bernoulli}\left(p^{\mathrm{net}}_{k,r} \right), \tag{194}$$

and the successful transmission delay follows

$$T^{\mathrm{net}}_{k,r} \sim \mathrm{LogNormal}\left(\mu^{\mathrm{net}}_{k,r}, \sigma^2_{\mathrm{net}} \right). \tag{195}$$

The downloaded version is generated before the current attempt:

$$s_{k,r} = r - A^{\mathrm{start}}_{k,r}, \qquad A^{\mathrm{start}}_{k,r} \geq 0, \tag{196}$$

where $A^{\mathrm{start}}_{k,r}$ is the model age at attempt registration. The staleness at window closure is

$$\tau_{k,r} = r - s_{k,r} = A^{\mathrm{start}}_{k,r}. \tag{197}$$

The arrival time is

$$T^{\mathrm{arr}}_{k,r} = T^{\mathrm{reg}}_{k,r} + T^{\mathrm{cmp}}_{k,r} + T^{\mathrm{net}}_{k,r}. \tag{198}$$

The usable-update indicator is

$$U_{k,r} = \mathbf{1}\Big\{ C^{\mathrm{cmp}}_{k,r} = 1,\ C^{\mathrm{net}}_{k,r} = 1,\ T^{\mathrm{arr}}_{k,r} < T_{r+1},\ \tau_{k,r} \leq S_{\max} \Big\}.$$

All usable arrivals in the same window are aggregated jointly. No individual arrival triggers an intermediate global update.

The practical usable-update model uses only information available at registration, including model age, expected computation time, expected transmission delay, device capability, energy, bandwidth, corrected local mass, and historical availability.

For the controlled simulator, a high-accuracy diagnostic propensity is obtained through conditional Monte Carlo:

$$\widehat{q}_{k,r}^{\mathrm{sim}} = \frac{1}{M_{\mathrm{MC}}} \sum_{m=1}^{M_{\mathrm{MC}}} U_{k,r}^{(m)}, \tag{199}$$

where each replay fixes the pre-attempt state and resamples only computation and network outcomes. The method using this propensity is denoted **RAVEN-SimOracle**, rather than claiming access to a closed-form true propensity.

F. Selection Regimes

We consider the following regimes.

1) **Balanced**: the opportunity distribution matches the target, and both selection propensities are approximately constant;
2) **Opportunity-only**: the recurring opportunity distribution differs from the target, while both selection stages are balanced;
3) **Observation-only**: opportunities match the target, but valid local observations are non-random;
4) **Usable-only**: opportunity and observation processes are balanced, but computation, transmission, and expiry are non-random;
5) **Complete**: opportunity imbalance and both non-random selection stages occur simultaneously;
6) **Drifting**: the complete regime is combined with a time-varying opportunity distribution.

The complete regime is used for the principal comparison. We vary the mean observation rate, mean usable-update rate, opportunity-skew intensity, drift intensity, client population, aggregation-window duration, and maximum accepted staleness.

G. Support and Reachability Audit

Formal positivity alone is insufficient when a target stratum has negligible effective opportunity mass. For every target client–stratum pair, define its expected effective opportunity count:

$$N_{k,s}^{\mathrm{eff}} = \sum_{r} \sum_{i \in \mathcal{R}_{k,r}: s(i)=s} p_{k,r,i}^{\mathrm{obs}} q_{k,r}^{\mathrm{use}}. \tag{200}$$

The group-level effective count is

$$N_g^{\mathrm{eff}} = \sum_{k,s} N_{k,s}^{\mathrm{eff}} \mathbf{1}\left\{ s \text{ overlaps } g \right\}. \tag{201}$$

Primary experiments require

$$N_g^{\mathrm{eff}} \geq N_{\min}^{\mathrm{eff}}, \qquad \forall g : \mu_g > 0. \tag{202}$$

We additionally report the lower quantiles of p^{obs}, q^{use}, and π^{opp} over the target support.

For a block $\mathcal{B}$ of active aggregation windows, empirical reachability is

$$\widehat{\epsilon}_{\mathrm{reach}}(\mathcal{B}) = \min_{\boldsymbol{\alpha}_r \in \mathcal{C}_r} \frac{\left\| \sum_{r \in \mathcal{B}} \eta_r \left(\boldsymbol{M}_r \boldsymbol{\alpha}_r - \boldsymbol{\mu} \right) \right\|_2}{\sum_{r \in \mathcal{B}} \eta_r}. \tag{203}$$

Unsupported and poorly reachable cases are reported as stress tests and are not mixed with the primary theory-consistent results.

H. Compared Methods

The principal federated comparisons use the same completion backbone, public field, client partitions, observation opportunities, local training budget, initialization, and aggregation windows.

- **Central-Delivered**: centrally trains on raw observations belonging to usable updates. It has the same realized information reachability as federated methods but does not preserve raw-data locality.
- **Central-All**: centrally trains on all valid local observations before computation and delivery loss. It is an information-rich reference rather than a guaranteed upper bound on target-weighted accuracy.
- **FedAvg-Window**: performs sample-count-weighted aggregation over each usable window without staleness or selection correction.
- **FedAsync-Window**: uses conventional sample-mass and staleness weighting over the usable set.
- **TimeAlign-Agg**: applies timestamp-aligned asynchronous aggregation to the same completion backbone.
- **FLAMF-Original**: reproduces the original time-aware completion and asynchronous aggregation pipeline. Because its completion backbone differs, it is treated as an external system baseline.
- **Local-Hajek**: applies opportunity-design and observation correction but no usable-update correction or target calibration.
- **TwoStage-Hajek**: applies both stages of correction and directly sets $\boldsymbol{\alpha}_r = \boldsymbol{\beta}_r$.
- **Inst-Cal**: uses instantaneous group calibration but no target debt.
- **Debt-Cal**: uses target debt and instantaneous group calibration without opportunity or propensity correction.
- **RAVEN-MCS**: uses the complete proposed framework with lagged estimated design ratios and propensities.
- **RAVEN-SimOracle**: uses the generating opportunity ratios, observation propensities, and Monte Carlo usable-update propensities in controlled experiments.

The principal experiments use local SGD, consistent with the theoretical analysis. Adaptive local optimizers are evaluated separately as an engineering robustness test.

I. Evaluation Metrics

1) Target-weighted completion accuracy: Let $\mathcal{I}^{\mathrm{te}}$ be the supported test set. The target weights are renormalized as

$$\widetilde{\varpi}_i^{\mathrm{tar}} = \frac{\varpi_i^{\mathrm{tar}}}{\sum_{\ell \in \mathcal{I}^{\mathrm{te}}} \varpi_\ell^{\mathrm{tar}}}. \tag{204}$$

The target-weighted MSE is

$$\mathrm{MSE}_\mu = \sum_{i\in\mathcal{I}^{\mathrm{te}}} \widetilde{\varpi}_i^{\mathrm{tar}} \left(\widehat{Y}_i - Y_i\right)^2, \tag{205}$$

and the primary metric is

$$\mathrm{RMSE}_\mu = \sqrt{\mathrm{MSE}_\mu}. \tag{206}$$

Target-weighted MAE is defined analogously.

Groups are ranked using the ratio of their raw arrival mass to target mass:

$$R_g^{\mathrm{arr}} = \frac{\widehat{\rho}_g^{\mathrm{arr}}}{\mu_g}. \tag{207}$$

The bottom and top 20% form the tail and head sets. We report tail RMSE and

$$\mathrm{Gap}_{\mathrm{HT}} = \mathrm{RMSE}_{\mathrm{tail}} - \mathrm{RMSE}_{\mathrm{head}}. \tag{208}$$

2) Raw arrival distribution and arrival-weighted risk: The realized raw client–record arrival distribution during training is

$$\widehat{\rho}_{k,i}^{\mathrm{arr,tr}} = \frac{\sum_r U_{k,r} O_{k,r,i} \mathbf{1}\left\{i \in \mathcal{R}_{k,r}\right\}}{\sum_{j,\ell,r} U_{j,r} O_{j,r,\ell} \mathbf{1}\left\{\ell \in \mathcal{R}_{j,r}\right\}}. \tag{209}$$

Its group marginal is

$$\widehat{\rho}_g^{\mathrm{arr}} = \sum_k \sum_{i:h(i)=g} \widehat{\rho}_{k,i}^{\mathrm{arr,tr}}. \tag{210}$$

Because test atomic units occur after the training period, their arrival weights are obtained from the fitted recurring opportunity and selection models. For test unit i, define the predicted raw arrival intensity

$$\begin{aligned}\widehat{r}_i^{\mathrm{arr}} = \sum_k \widehat{\pi}_{k,s(i)}^{\mathrm{opp}} \widehat{\nu}_k^{\mathrm{opp}}(i \mid s(i)) \\ \times \widehat{p}_{k,i}^{\mathrm{obs}} \widehat{q}_{k,s(i)}^{\mathrm{use}}.\end{aligned}$$

The normalized test arrival weight is

$$\widehat{\rho}_i^{\mathrm{arr}} = \frac{\widehat{r}_i^{\mathrm{arr}}}{\sum_{\ell\in\mathcal{I}^{\mathrm{te}}} \widehat{r}_\ell^{\mathrm{arr}}}. \tag{211}$$

The arrival-weighted completion error is

$$\boxed{\mathrm{RMSE}_\rho = \sqrt{\sum_{i\in\mathcal{I}^{\mathrm{te}}} \widehat{\rho}_i^{\mathrm{arr}} \left(\widehat{Y}_i - Y_i\right)^2}.} \tag{212}$$

Unlike a group-only metric, (212) also captures within-group atomic-unit arrival imbalance.

The target–arrival error gap is

$$\mathrm{Gap}_{\mathrm{mis}} = \mathrm{RMSE}_\mu - \mathrm{RMSE}_\rho. \tag{213}$$

3) Effective training distribution: Let $w_{k,r,i}^{(m)}$ be the normalized expected local-objective weight used by method m for record i in window r. It is zero when record i is not included in the current local objective.

For RAVEN-MCS,

$$w_{k,r,i}^{(\mathrm{RAVEN})} = \overline{a}_{k,r,i}. \tag{214}$$

For an uncorrected baseline, it is the normalized raw sample weight.

The learning-rate-weighted effective training distribution is

$$\widehat{\Pi}_{k,i}^{(m)} = \frac{\sum_{r=0}^{R-1} \eta_r I_r \alpha_{k,r}^{(m)} w_{k,r,i}^{(m)}}{S_R}. \tag{215}$$

It satisfies

$$\sum_{k,i} \widehat{\Pi}_{k,i}^{(m)} = 1. \tag{216}$$

For controlled experiments, we compare (215) with the target distribution renormalized over the training support:

$$\Delta_{\mathrm{pair}}^{(m)} = \left\| \widehat{\boldsymbol{\Pi}}^{(m)} - \widetilde{\boldsymbol{\pi}}^{\mathrm{tar,tr}} \right\|_1. \tag{217}$$

For large real traces, the primary estimable distribution metric is the client–stratum mismatch:

$$\Delta_{\text{c-s}}^{(m)} = \sum_{k,s} \left| \sum_{i:s(i)=s} \widehat{\Pi}_{k,i}^{(m)} - \pi_{k,s}^{\mathrm{tar}} \right|. \tag{218}$$

The effective group marginal is

$$\widehat{\omega}_g^{(m)} = \sum_k \sum_{i:h(i)=g} \widehat{\Pi}_{k,i}^{(m)}, \tag{219}$$

and its mismatch is

$$\Delta_{\mathrm{grp}}^{(m)} = \left\| \widehat{\boldsymbol{\omega}}^{(m)} - \boldsymbol{\mu} \right\|_1. \tag{220}$$

4) Per-window calibration diagnostics: The learning-rate-weighted pair-reference deviation is

$$\overline{\delta}_{\mathrm{ref}} = \frac{\sum_r \eta_r I_r \sqrt{K_r} \left\| \boldsymbol{\alpha}_r - \widehat{\boldsymbol{\beta}}_r \right\|_2}{S_R}. \tag{221}$$

The average instantaneous group discrepancy is

$$\overline{\delta}_{\mathrm{grp}} = \frac{\sum_r \eta_r I_r \left\| \boldsymbol{M}_r \boldsymbol{\alpha}_r - \boldsymbol{\mu} \right\|_1}{S_R}. \tag{222}$$

For controlled experiments, the realized per-window pair discrepancy is

$$\overline{\delta}_{\mathrm{pair}} = \frac{\sum_r \eta_r I_r \delta_r^{\mathrm{pair}}}{S_R}. \tag{223}$$

5) Debt and reachability: Normalized final debt is

$$\Delta_Q^{(R)} = \frac{\|\boldsymbol{Q}_R\|_1}{S_R}. \tag{224}$$

We jointly report

$$\|\overline{\boldsymbol{\varpi}}_R - \boldsymbol{\mu}\|_1 \quad \text{and} \quad \frac{2\|\boldsymbol{Q}_R\|_1}{S_R} \tag{225}$$

to verify Theorem 1.

We also report the empirical relationship among $\widehat{\epsilon}_{\text{reach}}$, debt growth, and final group mismatch.

6) Correction and system metrics: Opportunity-design estimation error is

$$\text{Err}_\zeta = \frac{1}{R}\sum_r \sum_{k,s} \pi^{\text{opp}}_{k,s,r} \left|\widehat{\zeta}_{k,r,s} - \zeta^\circ_{k,r,s}\right|. \tag{226}$$

Observation and usable-update propensity quality is measured using Brier score, log loss, and expected calibration error.

Stability metrics include local effective sample size, maximum aggregation weight, aggregate update variance, and seed-to-seed variation. System metrics include registration bytes, model-update bytes, client runtime, server calibration time, peak memory, and active windows required to reach a fixed target RMSE.

J. Implementation and Statistical Protocol

All same-backbone methods are implemented in PyTorch. The principal experiments use local SGD and identical client and server learning-rate schedules. Unless otherwise specified,

$$E_{\text{loc}} = 5. \tag{227}$$

We additionally evaluate

$$E_{\text{loc}} \in \{1, 5, 10, 30\} \tag{228}$$

to measure local-optimization drift.

The principal implementation solves **P2** using a deterministic convex solver with a relative tolerance of 10^{-6}. A projected first-order implementation is evaluated in the scalability experiment.

All clipping thresholds, forgetting factors, target-calibration coefficients, ESS constraints, variance coefficients, and staleness coefficients are selected using validation data. Every compared method receives the same hyperparameter-search budget.

Principal experiments use 20 independent random seeds. Large-scale sensitivity experiments use at least 10 seeds. We report mean, standard deviation, and 95% confidence intervals. Paired comparisons use a two-sided Wilcoxon signed-rank test with Holm correction.

K. Evidence of Target-Risk Misalignment

We first compare FedAvg-Window, FedAsync-Window, and TimeAlign-Agg under the balanced, opportunity-only, observation-only, usable-only, and complete regimes.

For each regime, we report

$$\begin{gathered}\Delta_{\text{grp}}, \quad \Delta_{\text{pair}} \text{ or } \Delta_{\text{c-s}}, \\ \text{RMSE}_\rho, \quad \text{RMSE}_\mu, \quad \text{Gap}_{\text{mis}}.\end{gathered} \tag{229}$$

Report whether arrival-weighted and target-weighted errors are statistically indistinguishable under the balanced regime.

Quantify the separate distributional distortions introduced by opportunity imbalance, observation selection, and usable-update selection.

Report whether conventional asynchronous methods retain relatively low arrival-weighted error while target-weighted and tail-group errors increase under the complete regime.

L. Overall Target-Risk Performance

Table II reports the principal results under the complete selection regime.

Report RAVEN-MCS's average relative target-RMSE improvement over the strongest deployable baseline.

Determine whether the improvement is concentrated in target-underrepresented units and groups.

Report the gap between RAVEN-MCS and RAVEN-SimOracle and relate it to opportunity and propensity-estimation errors.

Under the balanced regime, RAVEN-MCS should remain close to the strongest uncorrected method. A substantial degradation would indicate excessive importance-weight variance or over-aggressive calibration.

M. Ablation Study

We evaluate the following ablations:

- **w/o Design**: sets $\widehat{\zeta}_{k,r,i} = 1$;
- **w/o Obs**: removes inverse observation-propensity correction;
- **w/o Use**: removes inverse usable-update correction;
- **w/o Inst**: sets $\lambda_g = 0$;
- **w/o Debt**: removes the Lyapunov debt term;
- **w/o Ref**: sets $\lambda_\beta = 0$;
- **w/o Var**: removes the variance penalty and ESS constraint;
- **w/o Stale**: removes the staleness penalty;
- **No Forget**: estimates opportunity probabilities without temporal forgetting;
- **Current-Var**: replaces the lagged variance state with a same-update variance estimate.

The main text reports the first six ablations. Variance, staleness, opportunity drift, and same-update variance results may be placed in the appendix when space is limited.

Report the contribution of every component to target RMSE, pair mismatch, group mismatch, reference deviation, normalized debt, ESS, and maximum aggregation weight.

The contrast between **w/o Inst** and **w/o Debt** separates instantaneous target-gradient calibration from long-term group representation.

N. Robustness and Identification Boundary

We evaluate estimated correction under:

1) correctly specified opportunity and propensity models;
2) omitted observable opportunity features;
3) omitted observation-propensity features;
4) omitted usable-update features;

TABLE II: Target-risk performance under opportunity imbalance and two-stage non-random selection.

Method	$\mathrm{RMSE}_\mu \downarrow$	Tail RMSE $\downarrow$	$\Delta_{\mathrm{grp}} \downarrow$	$\Delta_{\mathrm{pair}} \downarrow$	$\overline{\delta}_{\mathrm{grp}} \downarrow$	$\overline{\delta}_{\mathrm{ref}} \downarrow$	$\Delta_Q^{(R)} \downarrow$
Central-Delivered	–	–	–	–	–	–	–
Central-All	–	–	–	–	–	–	–
FedAvg-Window	–	–	–	–	–	–	–
FedAsync-Window	–	–	–	–	–	–	–
TimeAlign-Agg	–	–	–	–	–	–	–
FLAMF-Original	–	–	–	–	–	–	–
Local-Hajek	–	–	–	–	–	–	–
TwoStage-Hajek	–	–	–	–	–	–	–
Inst-Cal	–	–	–	–	–	–	–
Debt-Cal	–	–	–	–	–	–	–
RAVEN-MCS	–	–	–	–	–	–	–
RAVEN-SimOracle	–	–	–	–	–	–	–

5) opportunity drift;
6) limited target support;
7) nonzero population-calibration residual; and
8) hidden confounding.

The hidden-confounding setting introduces a latent variable affecting both selection and the potential measurement or update. It intentionally violates (41) or (47) and is excluded from the main correctness claims.

Relate target error and pair mismatch to Err_ζ and propensity calibration error.

Measure how temporal forgetting affects performance as Γ_{drift} increases.

Report the target-risk residual produced by nonzero $\widehat{\Delta}_{\mathrm{cal}}$.

Show that hidden confounding produces persistent residual bias, confirming the stated identification boundary.

The clipping thresholds $a_{\max}$ and $d_{\max}$ are varied over a logarithmic grid.

Report the bias–variance tradeoff among target RMSE, distribution mismatch, effective sample size, maximum aggregation weight, and aggregate variance.

O. Theory Diagnostics

To verify Theorem 1, we plot

$$\|\overline{\boldsymbol{\omega}}_r - \boldsymbol{\mu}\|_1 \quad \text{and} \quad \frac{2\|\boldsymbol{Q}_r\|_1}{S_r} \tag{230}$$

throughout training.

Verify that measured long-term group mismatch never exceeds the deterministic debt-based upper bound.

We evaluate $\widehat{\epsilon}_{\mathrm{reach}}$ for several block lengths and compare it with debt growth and final group mismatch.

Report whether poorly reachable targets produce the residual group mismatch predicted by Theorem 2.

To examine the per-window gradient-bias chain, we jointly report

$$\overline{\delta}_{\mathrm{pair}}, \qquad \overline{\delta}_{\mathrm{grp}}, \qquad \overline{\delta}_{\mathrm{ref}}, \qquad \mathrm{RMSE}_\mu. \tag{231}$$

Determine whether pair-distribution discrepancy predicts target risk more accurately than final group proportions alone.

The lagged variance proxy is compared with repeated realizations of the same local update condition.

Report variance-proxy calibration, rank correlation, and the aggregate variance reduction obtained from the variance penalty.

P. Sensitivity and System Overhead

We vary:

$$K \in \{20, 50, 100, 200\}, \tag{232}$$

$$E_{\mathrm{loc}} \in \{1, 5, 10, 30\}, \tag{233}$$

$$S_{\max} \in \{1, 2, 5, 10, 20\}, \tag{234}$$

as well as aggregation-window duration, target-group resolution, opportunity-stratum resolution, warm-up size, heterogeneity level, and drift intensity.

Report whether RAVEN-MCS's advantage increases with opportunity skew, selection imbalance, and client heterogeneity.

Report how finer opportunity strata trade design-ratio bias against estimation variance.

Report the relationship among aggregation-window duration, usable-update count, staleness, reachability, and target error.

Report server optimization time for each client population and its fraction of total window runtime.

Report registration-message and group-summary communication overhead relative to TimeAlign-Agg.

Report the active windows and wall-clock time required to reach a fixed target RMSE.

RAVEN-MCS is considered practically scalable when calibration and summary-transmission overhead remain small relative to local model training and model-update transmission, while producing statistically significant reductions in target-weighted and tail-group completion errors.

VII. Conclusion and Future Work

This paper presented RAVEN-MCS, a target-risk-calibrated asynchronous federated data-completion framework for mobile crowdsensing under opportunity imbalance and two-stage non-random selection. RAVEN-MCS combines repeatable opportunity-stratum correction, Hájek-weighted local learning, usable-update propensity correction, instantaneous target calibration, and learning-rate-weighted target debt. Its strongly convex aggregation problem further controls corrected client composition, update variance, and model staleness. The theoretical analysis characterizes two complementary properties:

long-term group-mixture alignment under window reachability and target-risk convergence governed by the per-window client–record distribution error, local optimization drift, staleness, stochastic variance, and population-calibration residual. The evaluation protocol further distinguishes deployment-target risk from error measured under the raw arrival distribution.

RAVEN-MCS requires observable selection covariates, adequate target support, and sufficiently accurate opportunity and propensity models. Future work will investigate sensitivity-aware correction under hidden confounding, adaptive construction of opportunity strata under mobility drift, and privacy protection for transmitted participation summaries.

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